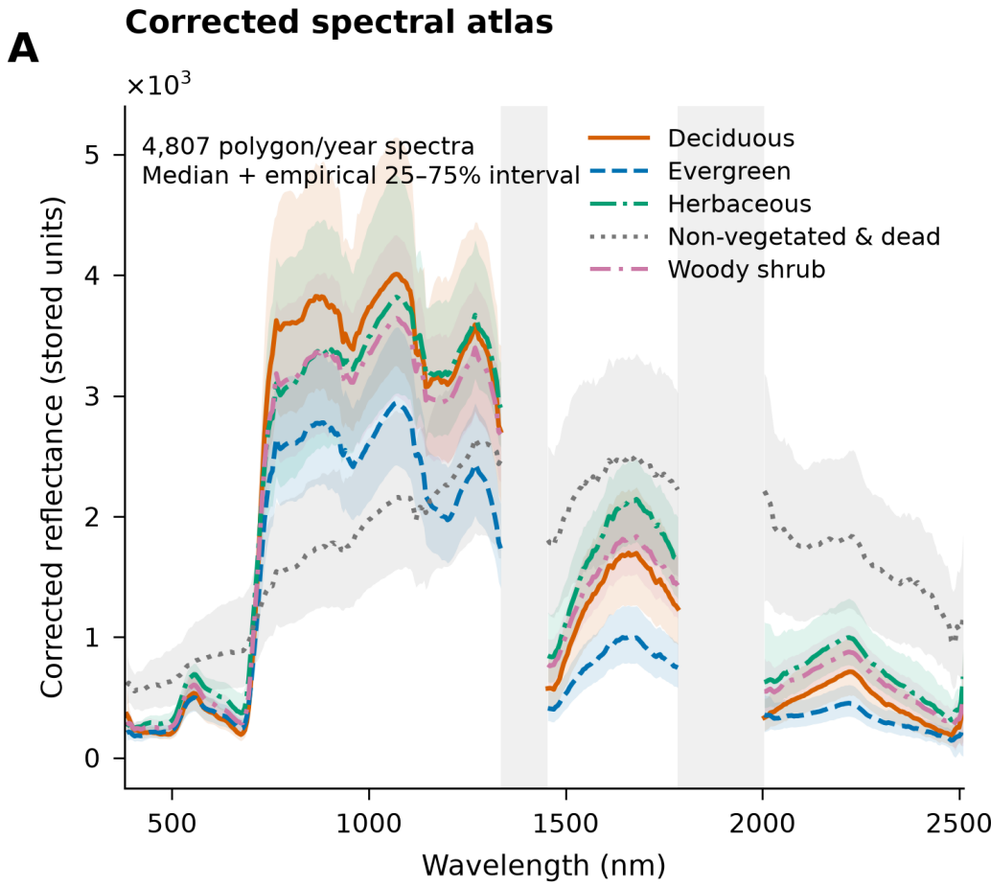
From spectral exploration to scientific readiness

What can the current spectral library distinguish, generalize, detect and unmix?



The central finding

The library contains useful biological spectral structure. Recognition transfers most reliably at broad woody leaf habit, while species, temporal transfer and mixture attribution are more fragile.

The scientific progression

We moved from trace plots and quantile signatures to matched whole-spectrum tests, grouped prediction, spectral reduction, an independent satellite pilot, and explicit tests of mixture ambiguity and uncertainty.

What this report is

A selection of 23 data-driven figure stories, with a reading guide and an explicit lesson for every figure. Existing experiments are reused; no models were refitted for this report.

For lab discussion | 15 October 2026

The cover shows category medians and empirical interquartile spread, not confidence bands. Values remain in source stored units. This PDF embeds every selected visual and can be shared on its own.

READER GUIDE

The story in seven steps

I. What is in the library? Figure 1 starts on p. 4

II. Explore before inferring Figure 2 starts on p. 5

III. Ask whether biological groups differ Figure 8 starts on p. 11

IV. Reduce spectral resolution, then test real imagery Figure 12 starts on p. 15

V. Predict on genuinely held-out data Figure 14 starts on p. 17

VI. Represent classes, detect mixtures, admit unknowns Figure 18 starts on p. 21

VII. Close the temporal and readiness questions Figure 22 starts on p. 25

Three distinctions to keep in mind

Difference: do sampled groups have different spectra?
Prediction: can an unseen labelled polygon be assigned correctly?
Unmixing: can the proportion of each material be recovered uniquely?
Evidence for one does not automatically establish the others.

The strongest current answer

Broad woody leaf habit: **96.7%** new-polygon and **90.1%** held-site NEON balanced accuracy. Species: **71.7%** and **47.4%**, respectively, conditional on supported targets.

Read the bottom of every figure page

HOW TO READ IT explains marks and axes. WHAT TO UNDERSTAND gives the scientific message. WHAT IT DOES NOT ESTABLISH sets the scope. Original panel letters are local to the embedded visual.

An evidence ladder, not a leaderboard. Early matching, later classification and synthetic fractions use different cohorts, metrics and targets. They are not interchangeable estimates of one accuracy. Methods: p. 3. Lab decisions: p. 27. Glossary: p. 28. Evidence guide: p. 29.

What the principal tests ask

Whole-spectrum permutation tests

We compare acquisition-matched group curves, then shuffle labels within the declared exchangeability blocks. A maximum statistic detects localized differences; an integrated statistic detects distributed ones. Their nonparametric combination (NPC) asks about the whole curve. Holm correction controls the declared comparison family.

Balanced classification accuracy

Recall is the fraction of each true class recognized. Balanced accuracy averages these recalls equally, so a common class cannot dominate the score. A two-class score has a different chance reference from a many-species score. Scores apply to eligible closed-set targets, whose coverage must also be reported.

Blocked holdouts and bootstrap intervals

All observations of an identity stay together. Site/year/acquisition holdouts ask specific transfer questions. Resampling whole IDs or spatial blocks preserves those clusters better than treating pixels as independent. Intervals remain conditional on the sampled domains and the stated design.

Spectral angle and spectral RMSE

Angle compares direction of spectral vectors and is less sensitive to a uniform brightness multiplier. Spectral RMSE measures magnitude mismatch in the stated units. Correlation measures co-variation, not equality. A good result for one metric need not be good for another.

Linear mixtures and fraction recovery

A controlled mixture combines parent spectra with known nonnegative fractions. We fit candidate spectra and compare estimated with known fractions. Multiple combinations can fit almost the same spectrum, so low spectral residual does not establish correct abundance. Here these are algebraic experiments without independent field fractions.

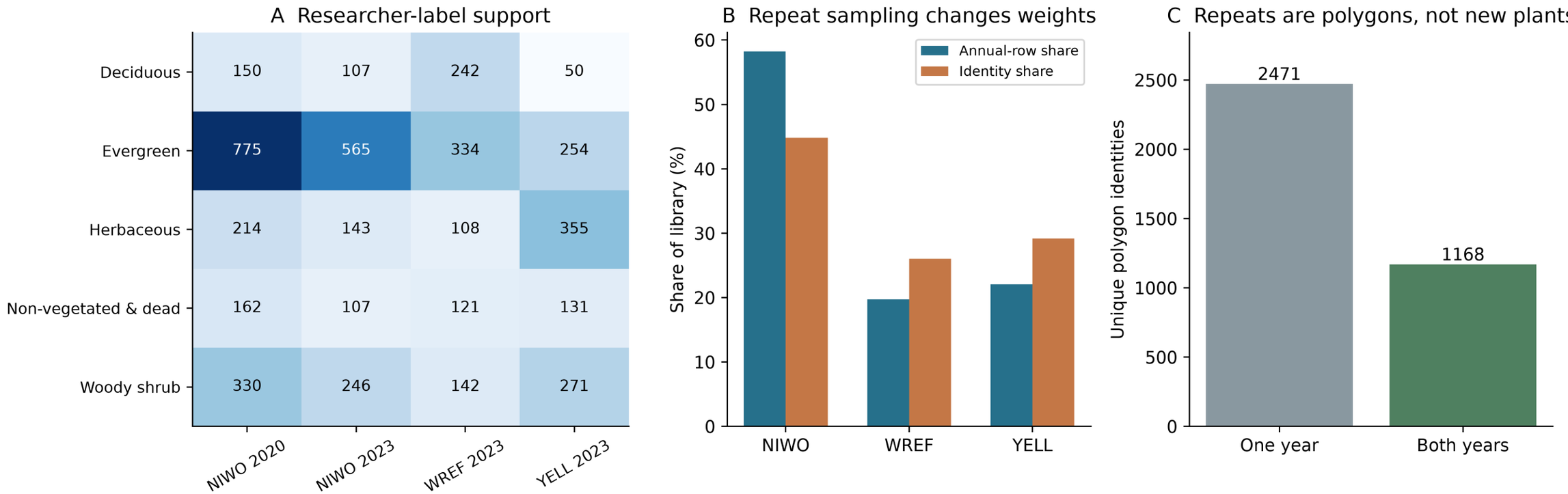
Calibration and prediction sets

Temperature scaling adjusts score sharpness on separate calibration identities. Conformal methods use a calibration-score quantile to return sets of possible labels. We measure whether the true label is in the set. Ordinary exchangeability-based coverage is not guaranteed after geographic shift.

Repeated-year partitions are descriptive sums of squares, not causal random-effect components. Source data are observational: labels, species, environment, illumination, acquisition and spatial support can remain confounded. Historical method definitions are retained in the evidence guide.

Figure 1. The sampling structure sets the limits of every result

What counts as replication, and which cohorts dominate?



HOW TO READ IT

A: category counts by site and image year. B: annual-spectrum share versus unique-polygon share. C: polygons observed in one or both years. NIWO, WREF and YELL are the three source-site codes.

WHAT TO UNDERSTAND

The library contains 85,706 extracted pixel records, 3,639 polygon identities and 4,807 polygon/year spectra. All 1,168 repeated-year identities are at NIWO. Counting annual rows equally increases NIWO weight: 58.2% of annual spectra versus 44.8% of unique identities.

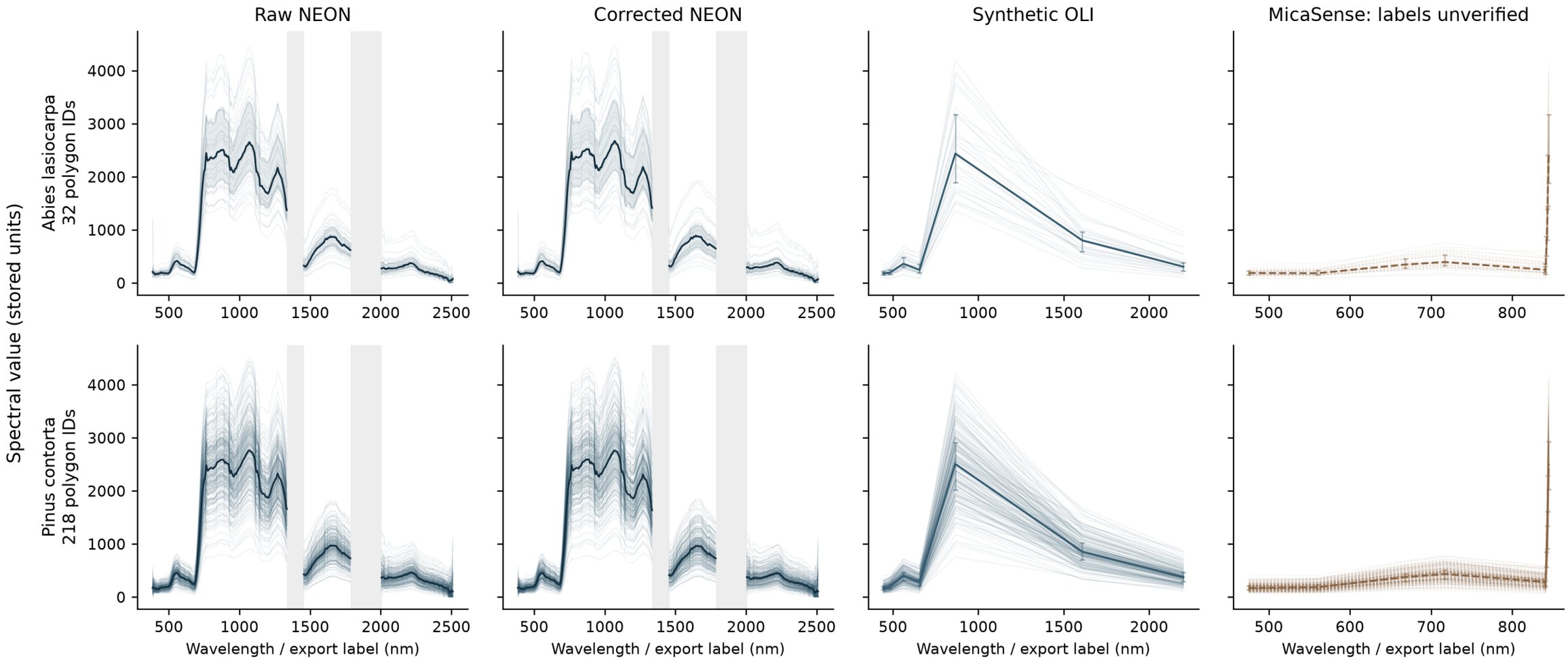
WHAT IT DOES NOT ESTABLISH

GlobalID identifies a polygon footprint, not a verified individual plant. Field annotations dated 2023/2024 were reused across image years. Native coordinate spacing is 1 m; median extracted support is two pixels per polygon/acquisition. Three sites cannot represent all regions.

Evidence: observational_support.csv; sampling_weight_effect.csv; annotation_year_audit.csv; native_spatial_support.csv. Full source paths and hashes are retained in the report packet.

Figure 2. Species have recognizable shapes, with substantial overlap

What did the first trace plots actually show?



HOW TO READ IT

Rows are selected named species; columns separate raw NEON, corrected NEON, derived OLI and archive MicaSense values. Thin curves show polygon summaries; the heavy curve is the displayed central summary. Use the original axis labels: these are stored spectral values, not verified unitless reflectance.

WHAT TO UNDERSTAND

The red-edge rise, high near-infrared signal and shortwave-infrared structure are visible, but spectra within a species span a broad range. Correction changes magnitudes; OLI retains only a few channels. This motivated distribution plots rather than one representative line per species.

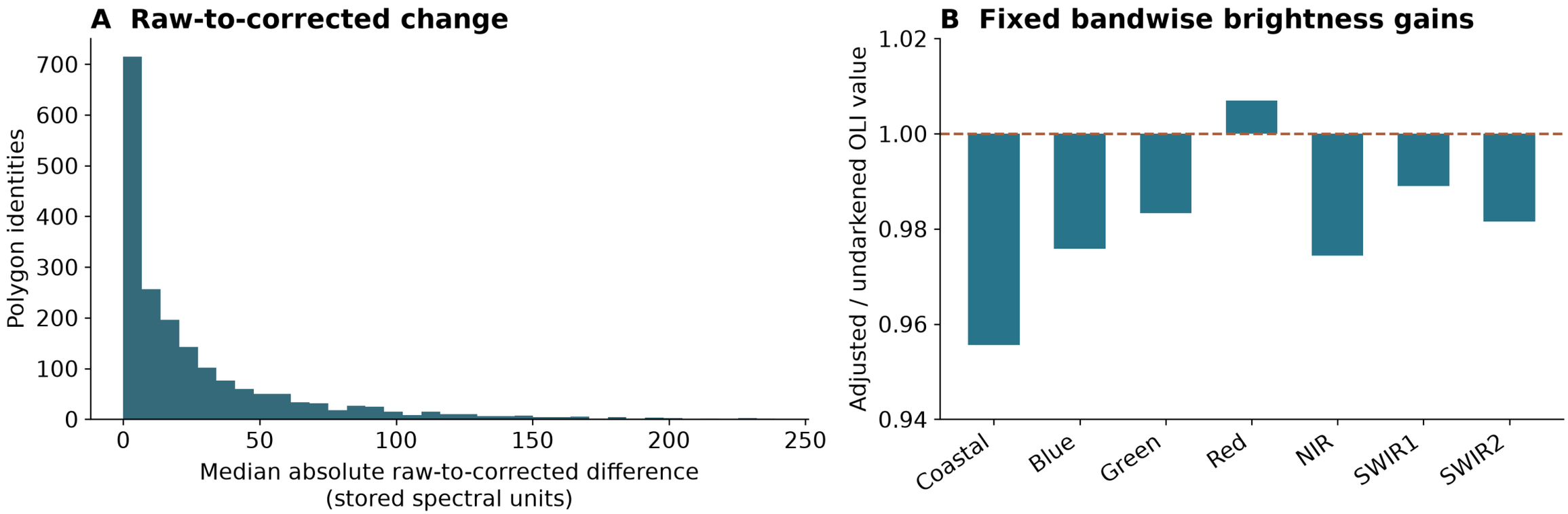
WHAT IT DOES NOT ESTABLISH

These are illustrative species, not the whole cohort. Traces join measured channels for readability and do not create extra bands. MicaSense coordinates are unverified: the sharp right-edge pattern cannot be interpreted biologically. Derived OLI and MicaSense are not independent satellite/drone observations.

Evidence: README.md; report_evidence.json. Full source paths and hashes are retained in the report packet.

Figure 3. Corrections and darkening are distinct operations

How much did processing change the observed values?



HOW TO READ IT

A: the histogram counts selected polygon identities by median absolute raw-to-corrected difference. B: saved per-band ratios show the OLI adjusted/undarkened gain; the dashed line is no change. Values below one reduce a band and values above one increase it.

WHAT TO UNDERSTAND

Most selected polygons have smaller correction differences, with a long tail. The operation called darkening is a fixed bandwise brightness adjustment, not a guarantee that every band gets darker. Its standardized matched group effects change by less than 8.8e-8, consistent with positive scale invariance.

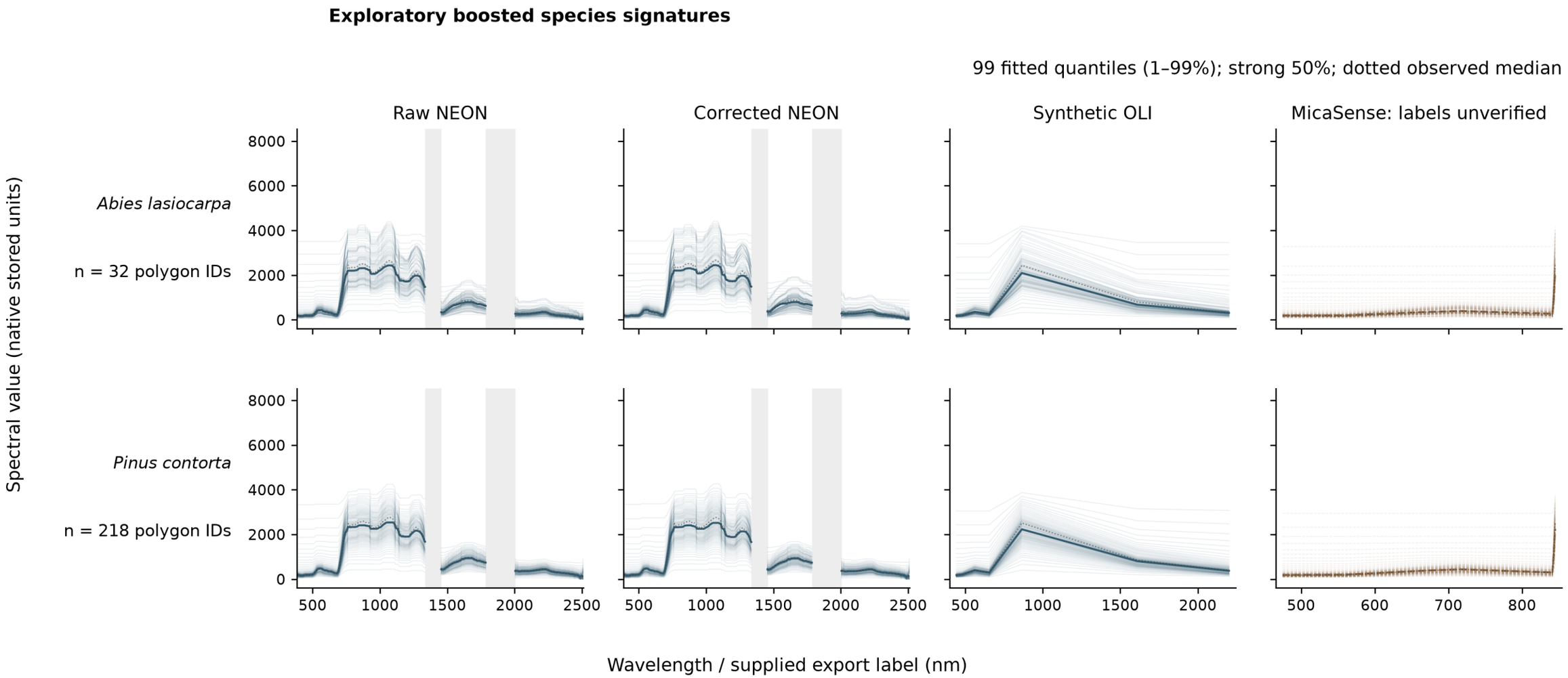
WHAT IT DOES NOT ESTABLISH

The histogram uses 1,886 selected labelled identities and stored units; it does not validate the physical correction. Scale-invariant tests cannot decide whether absolute radiometry improved. That question is addressed by the separate real-sensor pilot.

Evidence: report_evidence.json; brightness_audit.csv; RESULTS.md. Full source paths and hashes are retained in the report packet.

Figure 4. Quantile fans describe variation, not independent plants

Can a family of curves describe more than a species mean?



HOW TO READ IT

Rows are selected species; each sensor representation has its own column. Thin lines are boosted regressions for percentiles 1 through 99; the heavy line is the fitted median and the dotted reference is the empirical median. Gaps remain excluded rather than interpolated.

WHAT TO UNDERSTAND

We fitted both species-level and individual-polygon signatures. A fan can represent central and tail behavior and makes curve variability easier to compare. The final exploratory run used 99 quantiles, not an exact 100th percentile or 100 independent observations.

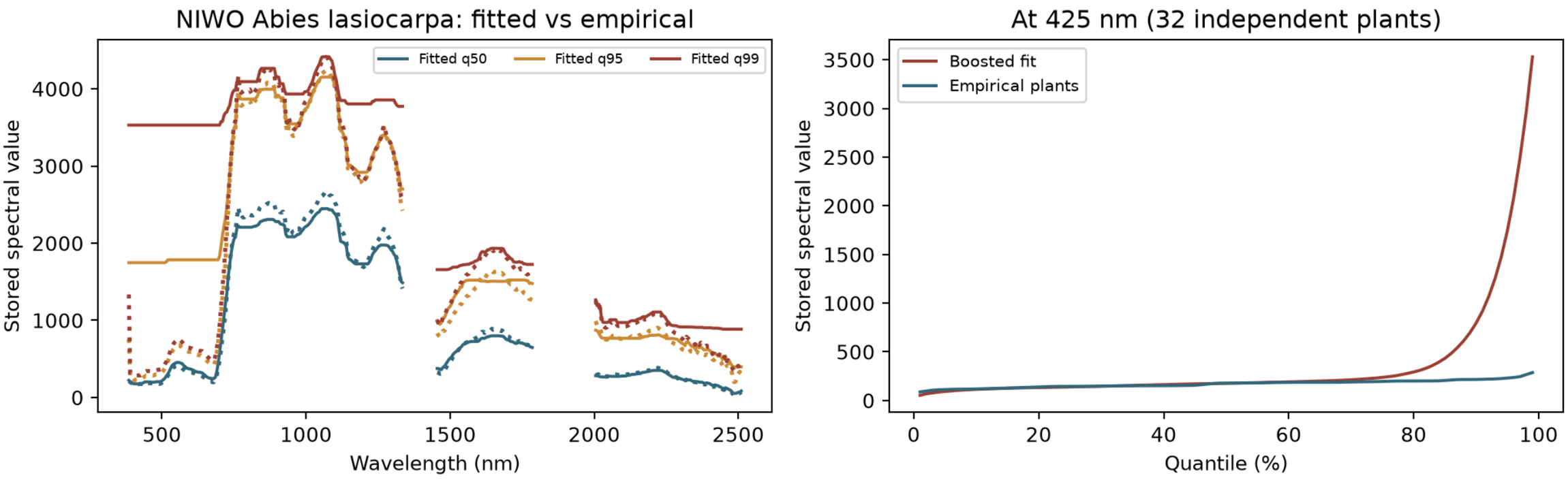
WHAT IT DOES NOT ESTABLISH

7,960 fitted signatures are model outputs, not the sample size. Individual signatures can have very little acquisition support; 62 selected polygons have only one observation. A fitted quantile fan is neither a confidence band nor a calibrated probability distribution over new plants.

Evidence: VALIDATION.md. Full source paths and hashes are retained in the report packet.

Figure 5. A dramatic upper tail was a fitting warning

Are the outer fitted curves faithful to the observations?



HOW TO READ IT

Left: fitted outer curves and empirical reference curves for NIWO Abies lasiocarpa. Right: at 425 nm, compare the fitted and empirical value at each percentile. The upper fitted curve diverges while the empirical tail remains much lower.

WHAT TO UNDERSTAND

At the 99th percentile and 425 nm, the fitted value is about 3,529 stored units versus an empirical value of about 287. The discrepancy is model behavior. It explains why a dense, visually impressive fan cannot be accepted without empirical checks.

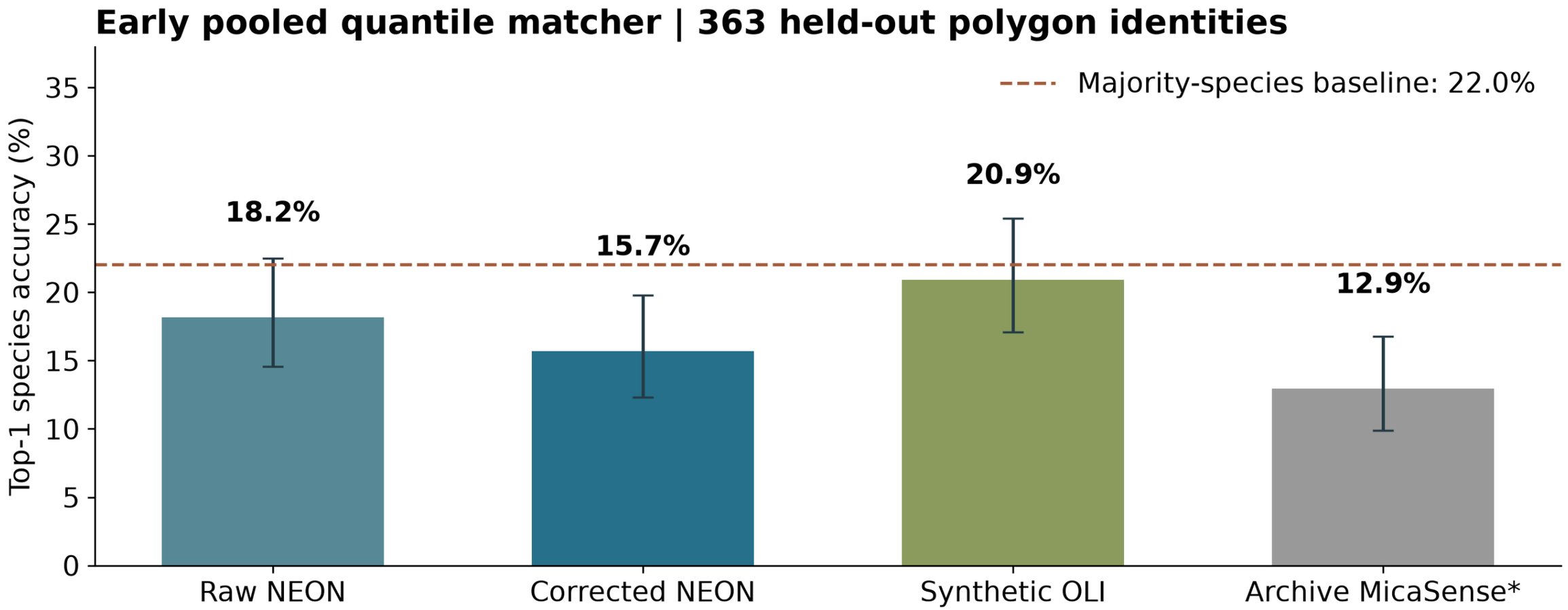
WHAT IT DOES NOT ESTABLISH

This is a selected failure diagnostic, not the error rate across all wavelengths/species. It does not prove all quantiles fail. The project did not adopt experimental tail-repair pilots as validated replacements.

Evidence: VALIDATION.md. Full source paths and hashes are retained in the report packet.

Figure 6. Simple quantile matching did not establish identification

Did the early curve-matching idea recognize held-out species?



HOW TO READ IT

Bars show ordinary top-1 species accuracy for 363 held-out polygon identities in the pooled early benchmark. Error bars are saved Wilson intervals. The dashed line is the majority-species baseline, not random equal-class chance.

WHAT TO UNDERSTAND

Corrected NEON achieved 15.7% and synthetic OLI 20.9%, below the 22.0% majority baseline. The earlier matching rule was not a useful general species identifier on this split. That motivated explicit class models, balanced metrics and structured holdouts.

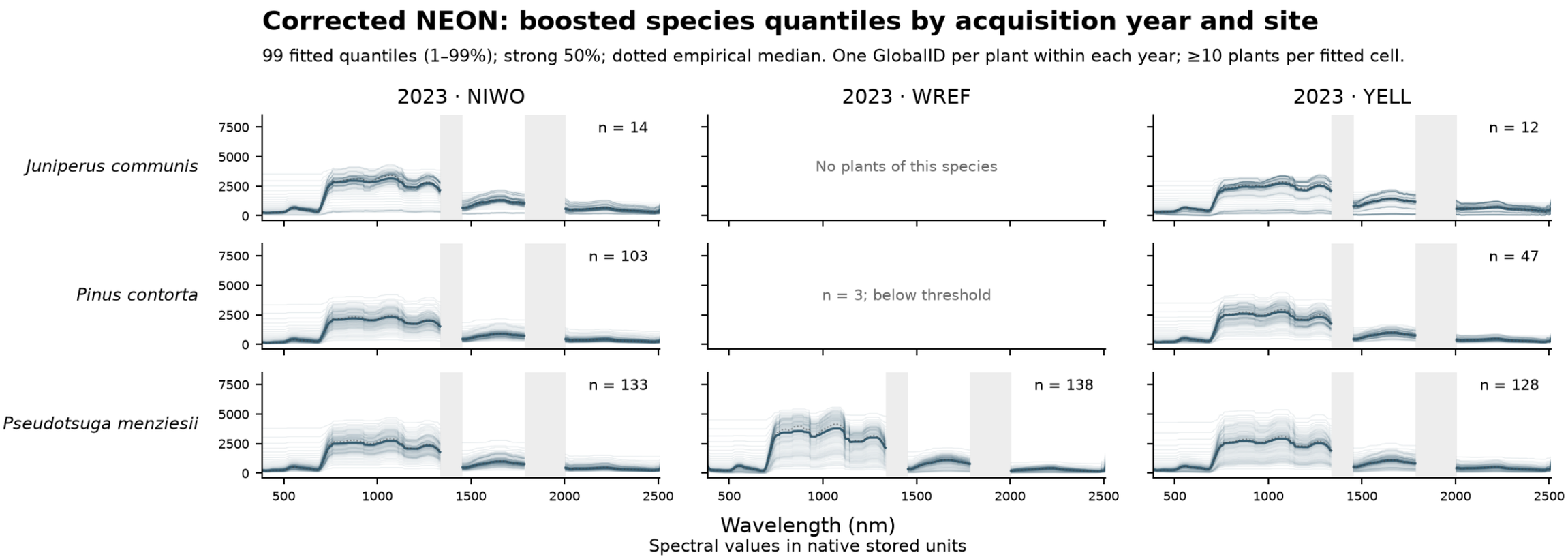
WHAT IT DOES NOT ESTABLISH

This benchmark uses a different task, class cohort, matcher and metric from the later logistic models. Do not subtract these accuracies from later balanced accuracy or attribute the difference to one processing change. MicaSense is shown only as an archived-label diagnostic.

Evidence: accuracy_intervals.csv; matching_baselines.csv. Full source paths and hashes are retained in the report packet.

Figure 7. Site-specific plots exposed an uneven comparison

Are the same species equally represented at all sites?



HOW TO READ IT

Columns are NIWO, WREF and YELL in image year 2023; rows are selected species. Read n in each panel and the explicit missing/low-support panels before comparing curve shapes. Lines are exploratory boosted quantiles, with an empirical median reference.

WHAT TO UNDERSTAND

Some species are absent or below the display threshold at a site, while others occur across all three. Geographic composition and spectral change are entangled. This is why a pooled species plot alone cannot answer regional transfer.

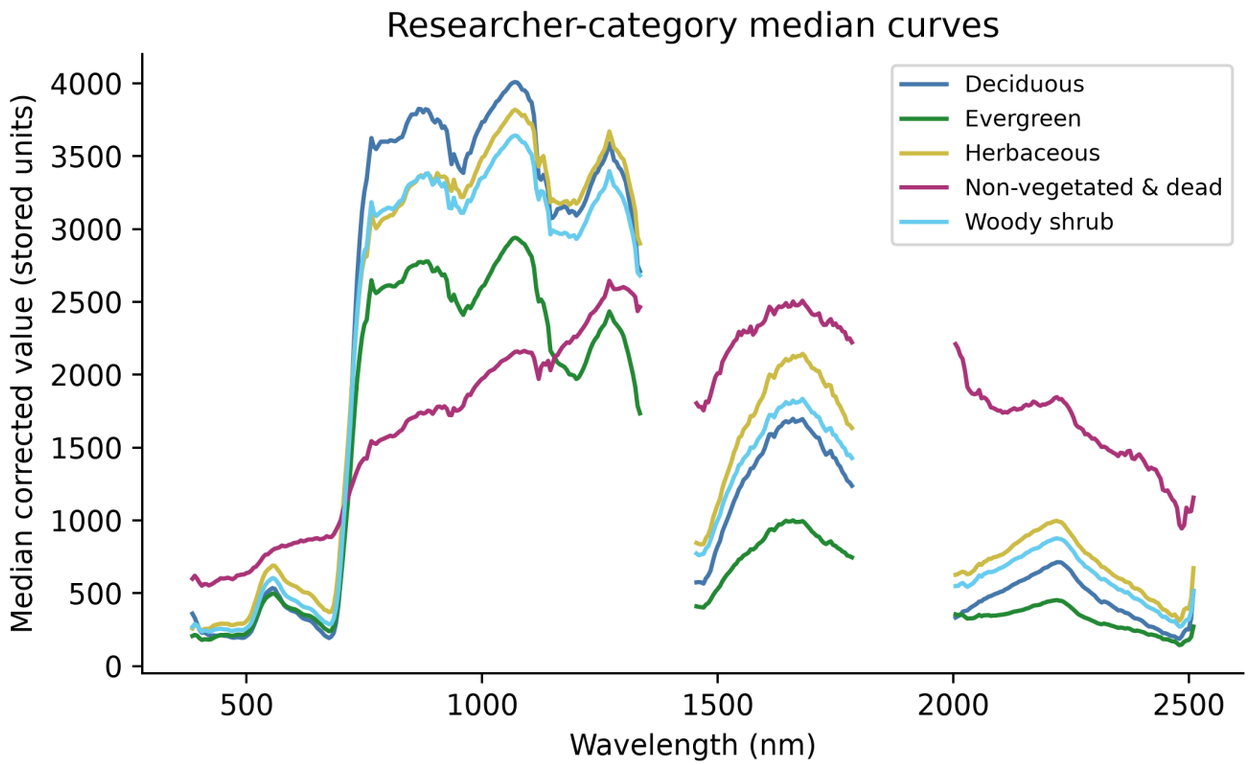
WHAT IT DOES NOT ESTABLISH

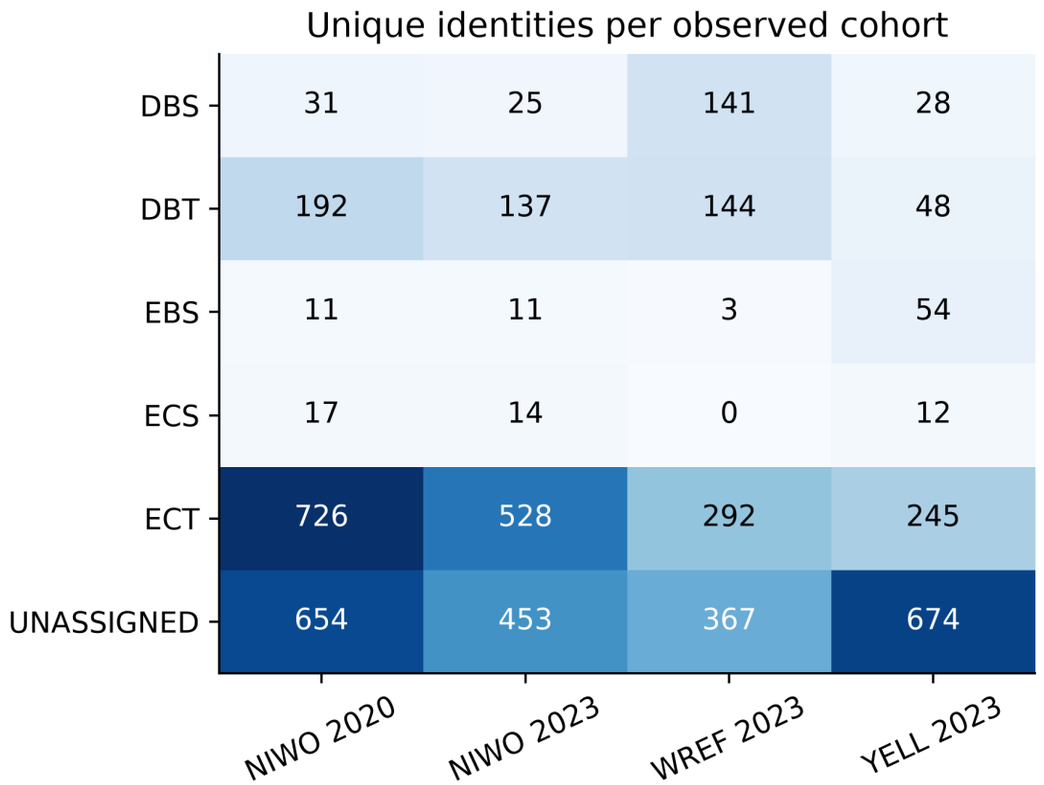
Displayed curve shifts may combine environment, acquisition and population composition, as well as fitting error. These selected panels are not a statistical site effect. Year-specific interpretation later uses complete paired NIWO identities and holds test histories out of training.

Evidence: README.md; VALIDATION.md. Full source paths and hashes are retained in the report packet.

Figure 8. There are two different five-group label systems

What does PFT mean in these analyses?





HOW TO READ IT

Left: full-cohort median spectra for the five researcher cover categories. Right: inferred woody-PFT identity counts by observed cohort. DBS/DBT are deciduous broadleaf shrub/tree; EBS is evergreen broadleaf shrub; ECS/ECT are evergreen conifer shrub/tree. UNASSIGNED is a missing inferred label, not a sixth PFT.

WHAT TO UNDERSTAND

Researcher categories use all 3,639 polygon IDs: Deciduous, Evergreen, Herbaceous, Non-vegetated & dead, Woody shrub. Species-derived woody PFTs use 1,944 assigned IDs. These populations answer different questions; the missing-species polygons are still useful for researcher-category analysis.

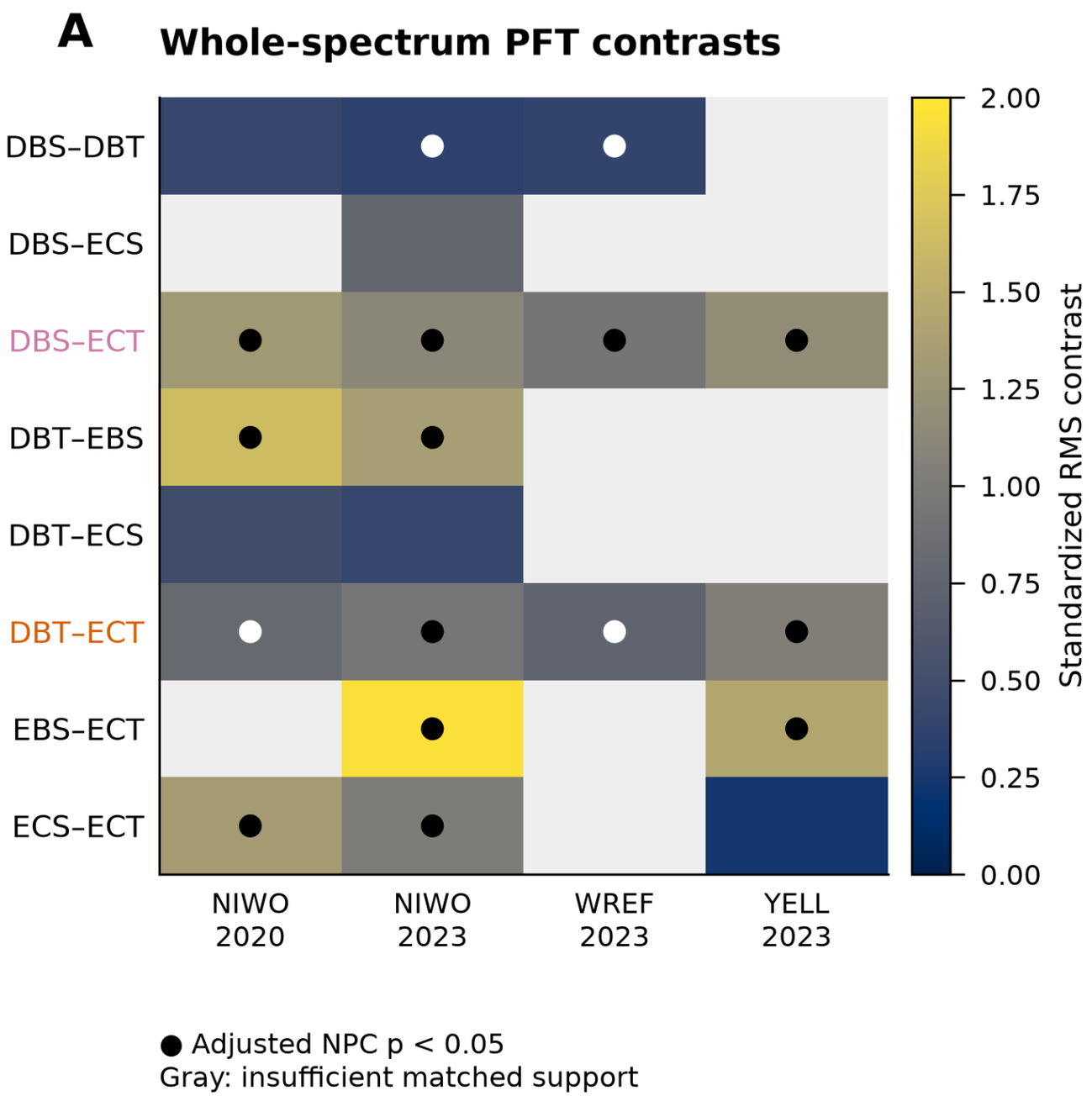
WHAT IT DOES NOT ESTABLISH

Non-vegetated & dead is heterogeneous, not a pure soil or dead-material endmember. A species-to-PFT lookup aliases species composition with traits. Neither label system is an independently measured causal physiological treatment. Full-cohort medians are not acquisition-matched contrast estimates.

Evidence: RESULTS.md; library_summary.csv. Full source paths and hashes are retained in the report packet.

Figure 9. Strong and weak woody-PFT contrasts coexist

Which inferred PFT pairs differ, within observed cohorts?



HOW TO READ IT

Rows compare PFT pairs; columns are site/year cohorts. Color is standardized root-mean-square curve contrast, not percent accuracy. Dots mark adjusted whole-spectrum test support. Blank cells lack matched support. Use the code definitions on the preceding page.

WHAT TO UNDERSTAND

Sixteen of 21 eligible NEON contrasts are supported in the fixed family. DBS versus ECT and DBT versus ECT are strong across cohorts; DBT versus ECS is weak in the NIWO years. Differentiation is contrast- and cohort-specific, not a universal statement that every PFT pair is separable.

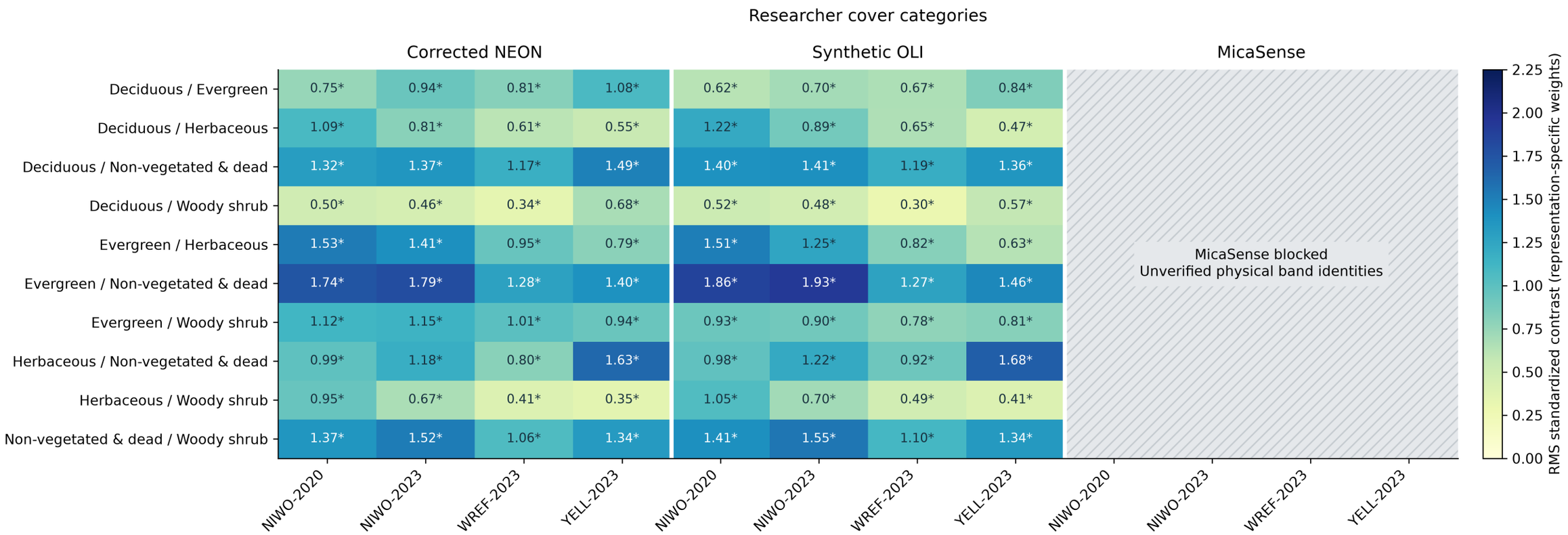
WHAT IT DOES NOT ESTABLISH

An acquisition-history-matched permutation test asks whether the sampled group curves differ under its exchangeability assumption. It does not establish nonoverlap, causal trait effects, equal distributions or reliable assignment of a new polygon. Species and site composition remain observational constraints.

Evidence: RESULTS.md; plotted_effects.csv. Full source paths and hashes are retained in the report packet.

Figure 10. The five researcher categories differ, by varying amounts

Does using the researchers' own category field change the answer?



* NPC Holm adjusted $p < 0.05$, family fixed within label system and sensor. White/blank = insufficient common support; hatched = metadata gate.
NEON: supported wavelength integral. OLI: equal channels. Values compare standardized separation, not literal information fractions.

HOW TO READ IT

Rows are the ten category pairs, with four site/year cohorts in each representation. Colors/numbers show standardized RMS separation; stars indicate Holm-adjusted combined-test support. Hatched MicaSense cells mean metadata-blocked, not a zero effect.

WHAT TO UNDERSTAND

All 40 eligible NEON comparisons and all 40 corresponding synthetic OLI comparisons are supported. NEON standardized effects span 0.336 to 1.795. Smaller differences include Deciduous versus Woody shrub at WREF; a uniform significance decision hides large variation in magnitude.

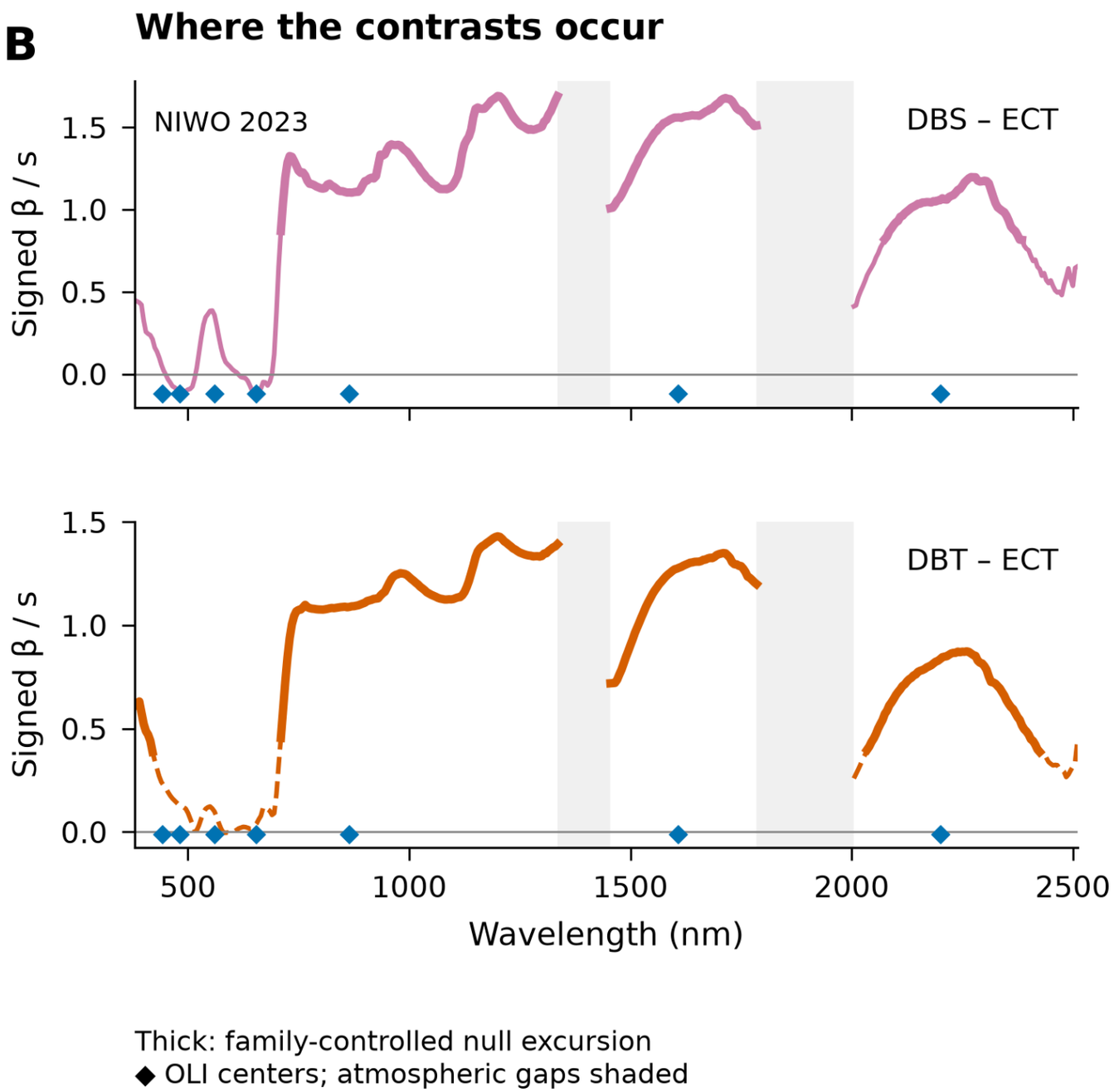
WHAT IT DOES NOT ESTABLISH

The combined test integrates maximum and distributed curve evidence, with family correction. Primary adjusted $p=0.004$ is a permutation-resolution floor, not identical effects. An energy-statistic sensitivity supports 38/40, so distribution-wide evidence is less consistent for two comparisons.

Evidence: RESULTS.md; methods_and_effects.csv; sensor_contrasts.csv. Full source paths and hashes are retained in the report packet.

Figure 11. Differentiation occupies broad spectral regions

Where along the spectrum do two strong contrasts occur?



HOW TO READ IT

The two NIWO 2023 examples plot signed adjusted group differences divided by pooled spectral SD. Positive values mean the first named group is higher. Thick segments mark family-controlled null exclusion; thin/dashed segments do not. Gray areas are atmospheric gaps; diamonds locate OLI bands.

WHAT TO UNDERSTAND

For these strong deciduous-versus-evergreen-tree contrasts, substantial differences extend across near-infrared and shortwave-infrared regions. This offers a spectral explanation for broad contrast retention after OLI convolution, while finer regions are averaged or absent.

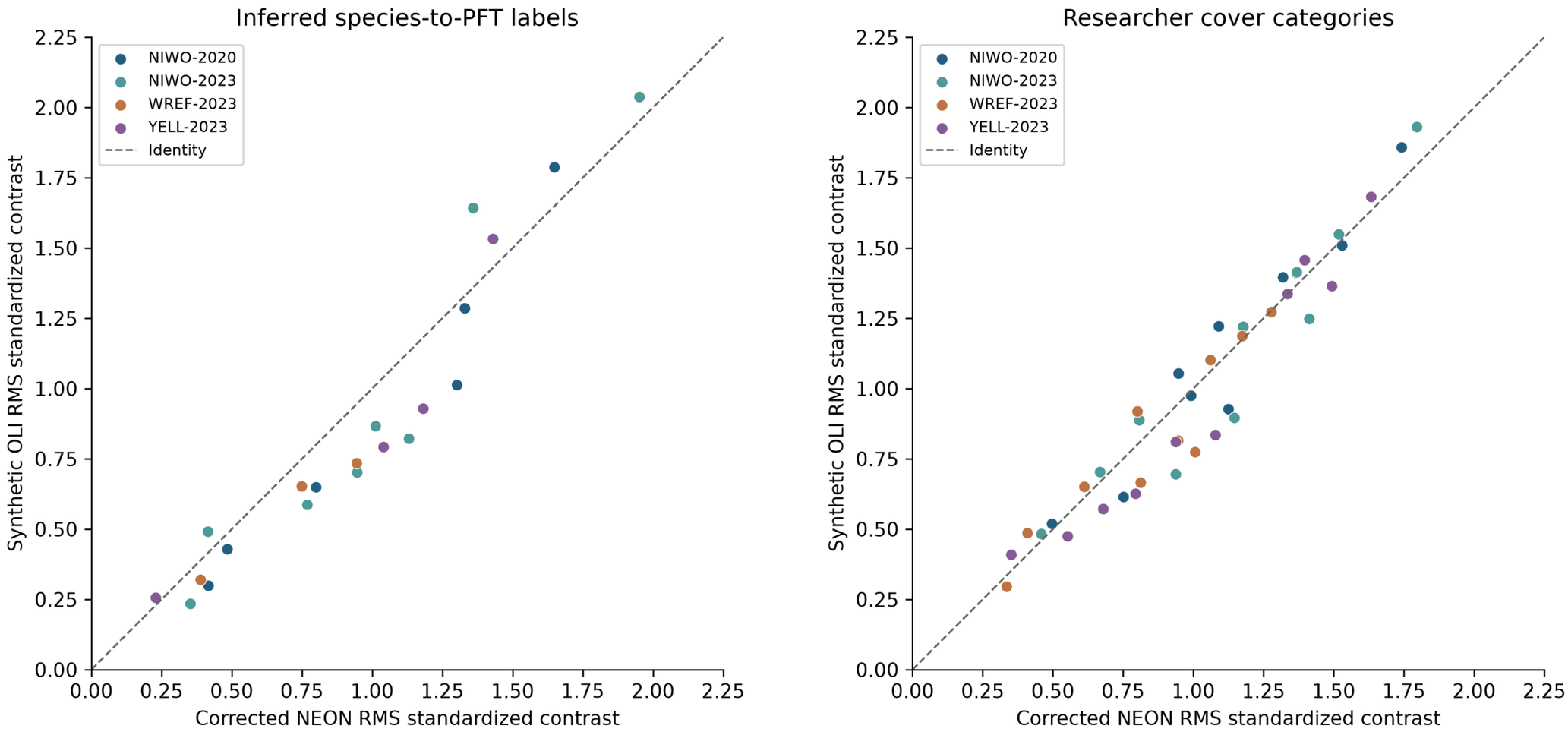
WHAT IT DOES NOT ESTABLISH

Localization is statistical evidence in these sampled curves, not a unique biochemical diagnosis. Correlated wavelengths can carry redundant information. Selected strong examples do not describe every pair, and no optimal universal band set follows from them.

Evidence: plotted_localization.csv; neon_region_coverage.csv. Full source paths and hashes are retained in the report packet.

Figure 12. Synthetic OLI retains much of the contrast structure

What survives seven-band spectral reduction on identical polygons?



Paired observations and histories throughout. Ratios use different declared spectral norms; values above identity do not establish sensor superiority.

HOW TO READ IT

Compare each paired NEON and OLI standardized effect; the identity reference means equal numerical effect. The two panels keep inferred PFTs and researcher categories separate. Samples and acquisition-history matching are identical across these sensor representations.

WHAT TO UNDERSTAND

OLI retains 15 of 16 NEON-supported inferred-PFT contrasts and all 40 researcher-category contrasts. The inferred-PFT median effect ratio is 0.824, with conditional paired-bootstrap interval 0.801-0.889. A borderline NIWO 2023 DBS/DBT contrast loses support.

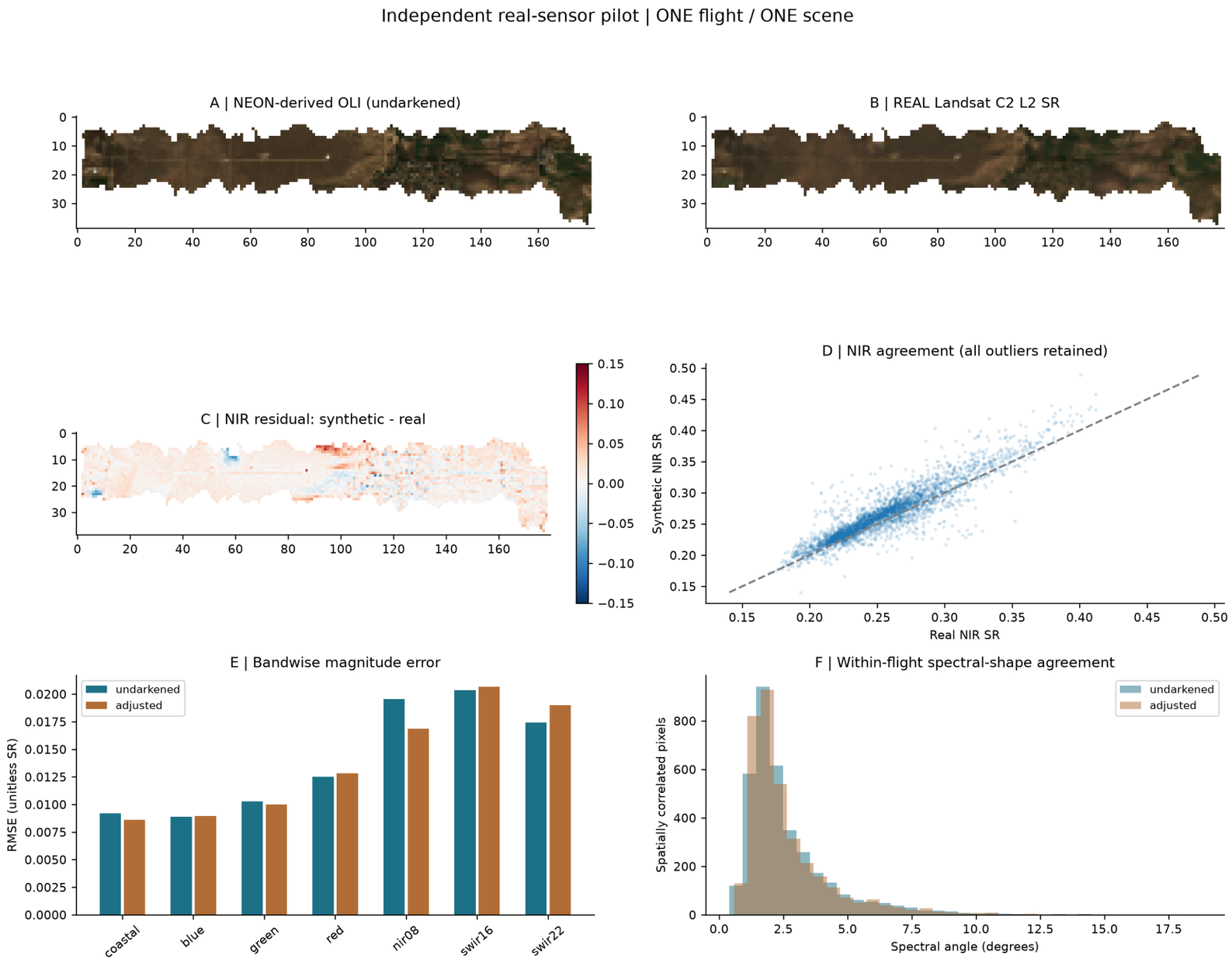
WHAT IT DOES NOT ESTABLISH

The 0.824 ratio is not 82.4% of all biological information: wavelength-integrated NEON and equal-channel OLI use different metrics. Shared-source convolution has no independent sensor noise or 30 m spatial mixing. MicaSense ecological inference remains blocked by unverified physical coordinates.

Evidence: effect_retention.csv; retention_summary.csv; RESULTS.md. Full source paths and hashes are retained in the report packet.

Figure 13. One real Landsat pilot is encouraging, with structured error

Does the synthetic spectrum agree with an independent satellite observation?



RGB shares extent/grid/mask and fixed 0-0.4 stretch. Primary uses original registration and no empirical calibration. Separate magnitude and shape diagnostics.
This single-flight pilot does not establish cross-flight/site replication, actual regional transfer, or an accuracy threshold for every scientific application.

HOW TO READ IT

A-B: matched image views. C: synthetic-minus-real near-infrared residual. D: NIR scatter around the identity line. E: bandwise error before/after brightness adjustment. F: spectral angles. SR denotes unitless surface reflectance; image axes are matched grid positions.

WHAT TO UNDERSTAND

At R10C on 15-16 September 2021, one NEON flight and one next-day Landsat 8 scene yield 3,583 matched 30 m footprints in 56 spatial bootstrap blocks. Pooled seven-band RMSE is 0.01477 and median spectral angle is 2.089 degrees. Broad agreement coexists with spatially and spectrally structured discrepancies.

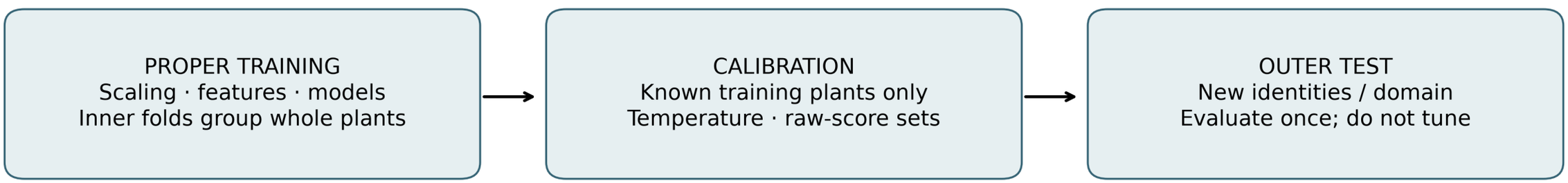
WHAT IT DOES NOT ESTABLISH

The lag is about 22.37 h conditional on the legacy timestamp basis. Overall brightness-adjustment RMSE change is -0.000184, with interval [-0.000440, +0.000041], spanning zero. This single acquisition pair validates a constrained comparison, not universal calibration or multi-site interchangeability.

Evidence: RESULTS.md; band_metrics.csv; real_landsat_flight_metrics.csv. Full source paths and hashes are retained in the report packet.

Figure 14. The evaluation target changes the scientific claim

What did we do to avoid training on the answer?



One GlobalID and all its years/flights stay in one role.

No test-guided feature selection, endmembers, tuning or thresholds.

HOW TO READ IT

Proper training learns scaling, features and model parameters. Calibration uses other training identities for temperature/threshold/set choices. The outer test evaluates unseen identities or a held-out acquisition, year or site. An entire GlobalID history stays in one role.

WHAT TO UNDERSTAND

A new-polygon test measures familiar-domain recognition. A held-site test asks a harder geographic-transfer question; a year test asks a temporal/domain question. Feature and endmember selection use training data only. Results are reported for supported class targets, not silently extrapolated to unseen labels.

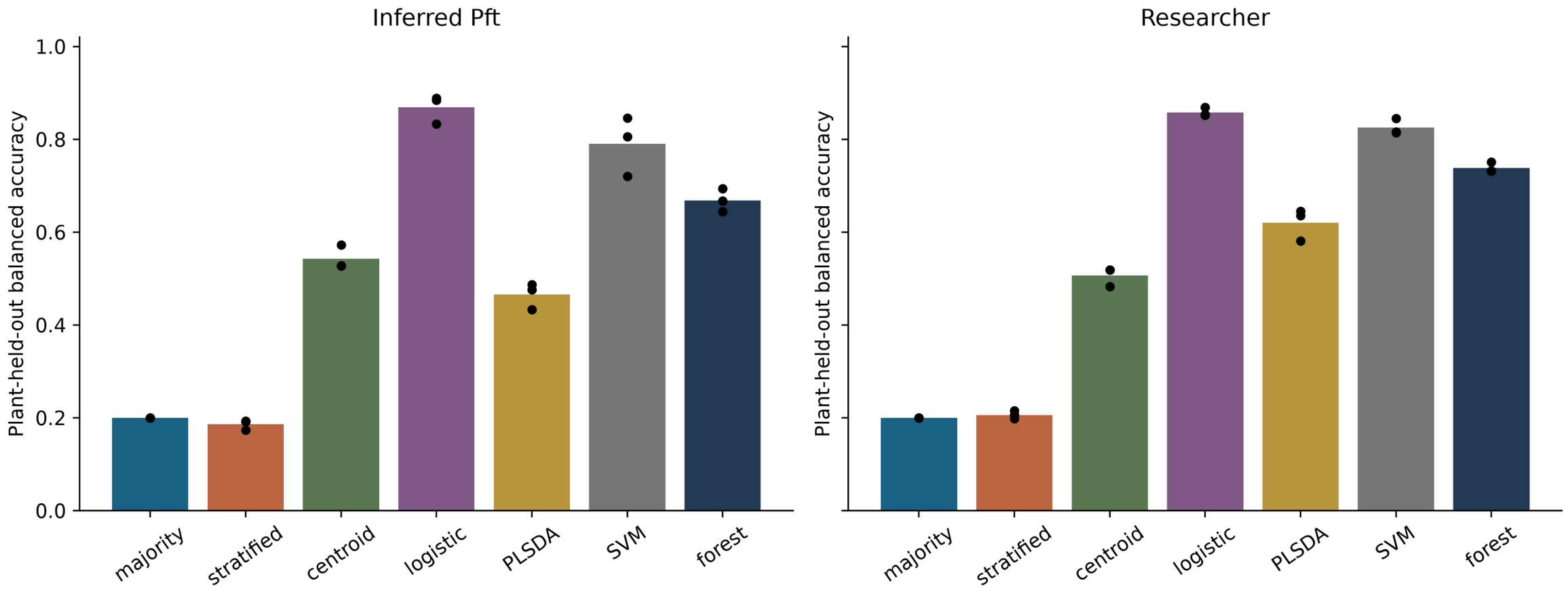
WHAT IT DOES NOT ESTABLISH

A polygon split is not complete ecological independence. Nearby footprints, shared species and environmental confounding can remain. Held-site tests have only three sites. Global year holdouts change site composition; the later paired NIWO test isolates the repeated-footprint comparison more directly.

Evidence: DESIGN.md; OBSERVATIONAL_STRUCTURE.md; numerical_validation.json. Full source paths and hashes are retained in the report packet.

Figure 15. More complicated classifiers were not consistently better

Do flexible models earn their added complexity?



HOW TO READ IT

Bars compare mean new-polygon balanced accuracy for two label systems. Dots show fold scores. Majority/stratified baselines precede centroid, logistic, partial-least-squares discrimination, support-vector and forest models. Balanced accuracy is mean recall across classes.

WHAT TO UNDERSTAND

On this task logistic regression is a strong reference, while several more elaborate alternatives do not consistently improve it. Class information is real, but model complexity alone is not the limiting factor. This justified focusing the final closure on interpretable, grouped logistic comparisons.

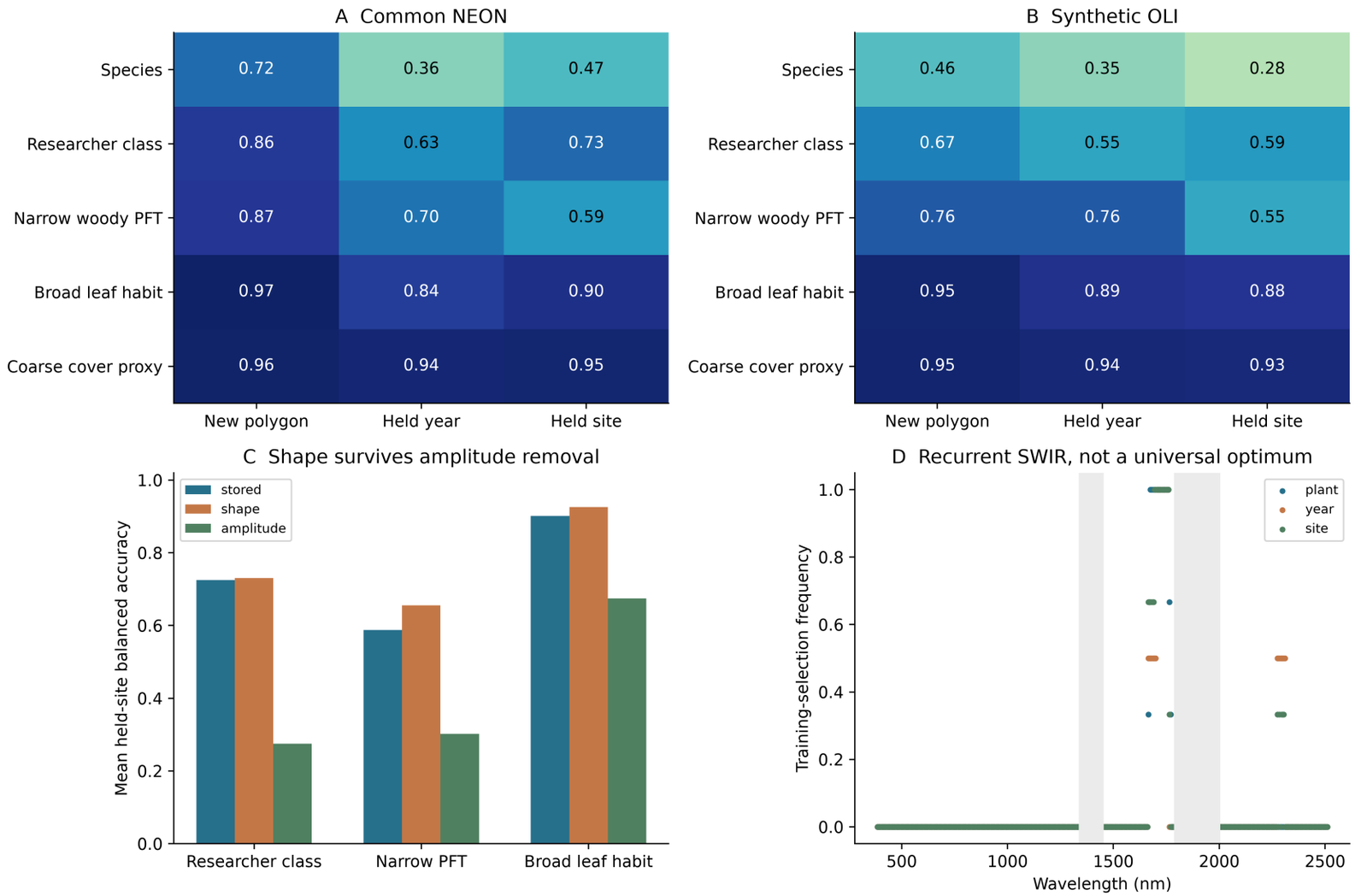
WHAT IT DOES NOT ESTABLISH

The figure is a familiar-domain model comparison, not a geographic transfer ranking. Fixed grids and available sample size constrain conclusions. These results do not establish that nonlinear models can never help, nor do they motivate a deep-model branch without new evidence.

Evidence: STAGE3_ML_GENERALIZATION.md; ml_generalization.csv. Full source paths and hashes are retained in the report packet.

Figure 16. Biological resolution governs how well recognition transfers

What can be recognized, at which label and sensor resolution?



HOW TO READ IT

A-B: mean fold balanced accuracy for supported labels under new-polygon, held-year and held-site tests, for common NEON and synthetic OLI. C: held-site stored spectra, unit-norm shape and one-amplitude models. D: frequency of training-selected coordinates; it is not a causal importance map.

WHAT TO UNDERSTAND

NEON new-polygon/held-site scores are 71.7/47.4% for species, 85.8/72.5% for researcher categories and 96.7/90.1% for broad woody leaf habit. Broad shape-only held-site accuracy is 92.6%, versus 67.4% for amplitude alone: recognition is not explained solely by brightness.

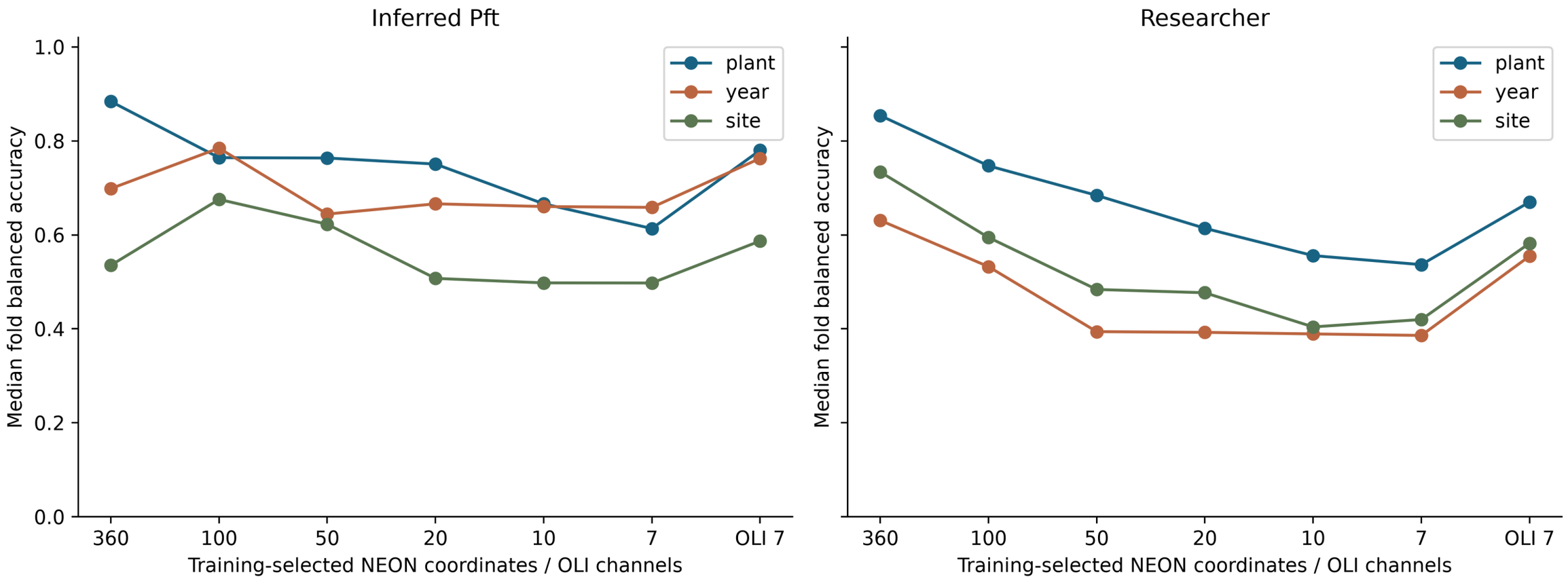
WHAT IT DOES NOT ESTABLISH

Broad leaf habit is a two-class deciduous/evergreen woody subset, not all five researcher classes. Label counts, chance and eligible populations differ. Species target coverage at WREF is only 24.3%; scores are conditional on supported labels. Normalization is a sensitivity, not a production change.

Evidence: biological_generalization.csv; closed_set_coverage.csv; brightness_sampling_sensitivity.csv. Full source paths and hashes are retained in the report packet.

Figure 17. A handful of neighboring bands is not an optimized sensor

Can a compact wavelength subset reproduce full-spectrum prediction?



HOW TO READ IT

Each curve links separately evaluated feature budgets; x is a discrete budget, not a continuous wavelength scale. Coordinates are selected using proper training only. The last OLI point is a different broad-response representation, not seven of the same narrow channels.

WHAT TO UNDERSTAND

Reducing channels changes prediction, and the outcome depends on transfer domain and label system. Seven selected narrow NEON coordinates yield about 63.2% new-polygon narrow-PFT balanced accuracy versus 76.4% for synthetic OLI. OLI can outperform this particular narrow-coordinate selection.

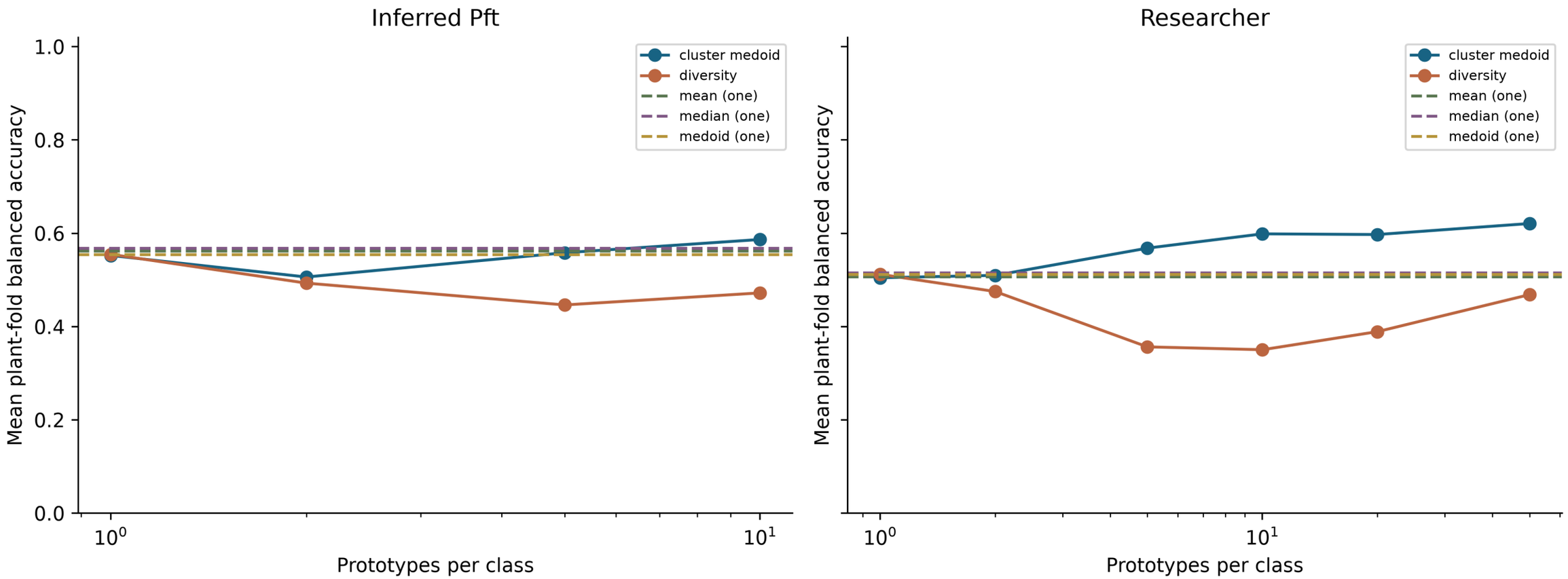
WHAT IT DOES NOT ESTABLISH

The narrow-PFT top-20 site selections concentrate at 1695-1760 nm, demonstrating repeated but highly correlated information. These experiments do not separate band number, placement and width; univariate selection is not a globally optimized sensor design.

Evidence: STAGE4_WAVELENGTH_INFORMATION.md; wavelength_selection.csv; spectral_resolution_performance.csv. Full source paths and hashes are retained in the report packet.

Figure 18. A single average does not span a heterogeneous class

How many reference spectra should represent each group?



HOW TO READ IT

The horizontal axis is prototype count per class. Curves compare observed cluster representatives, diversity-selected observations and central summaries. Dashed references indicate the plotted baseline. Only training spectra supply prototypes; outer tests remain held out.

WHAT TO UNDERSTAND

Several observed prototypes can improve coverage relative to one center, but gains depend on selection rule and label system. Increasing coverage does not guarantee correct class attribution: flexible libraries can also explain spectra using the wrong classes.

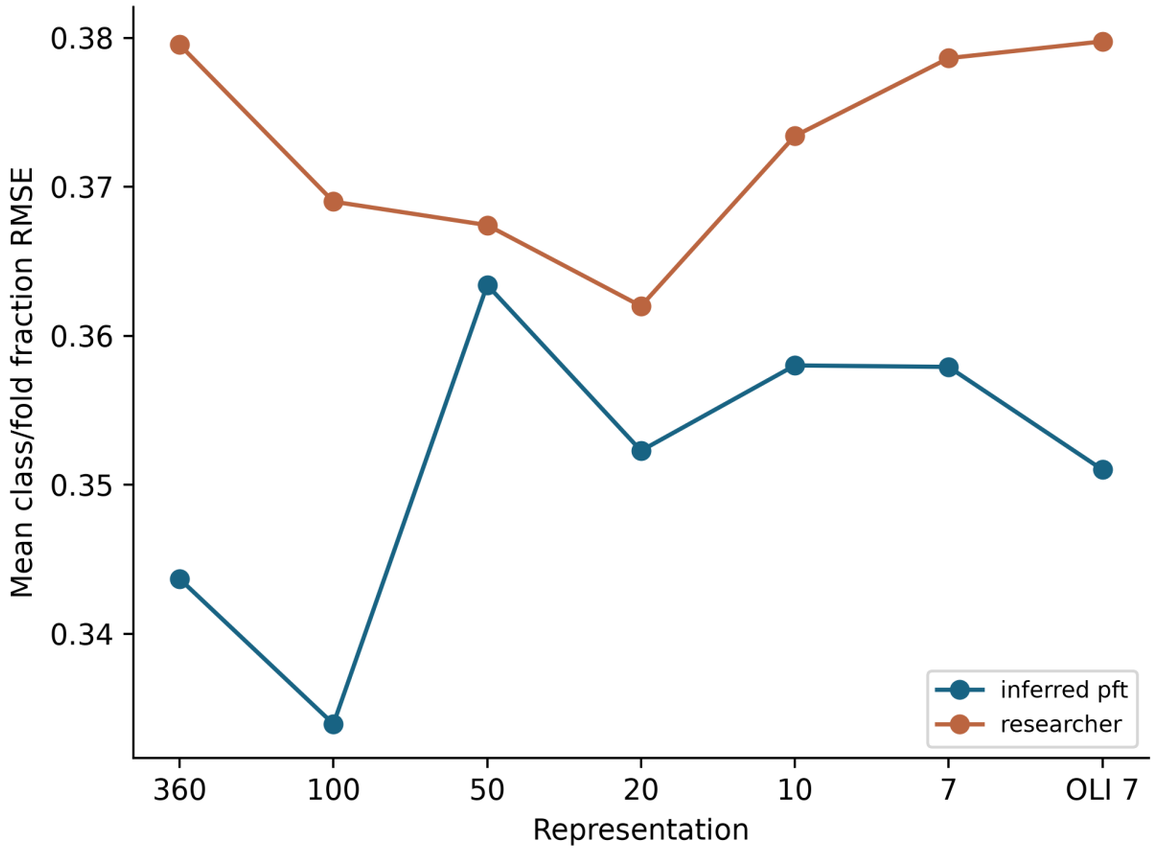
WHAT IT DOES NOT ESTABLISH

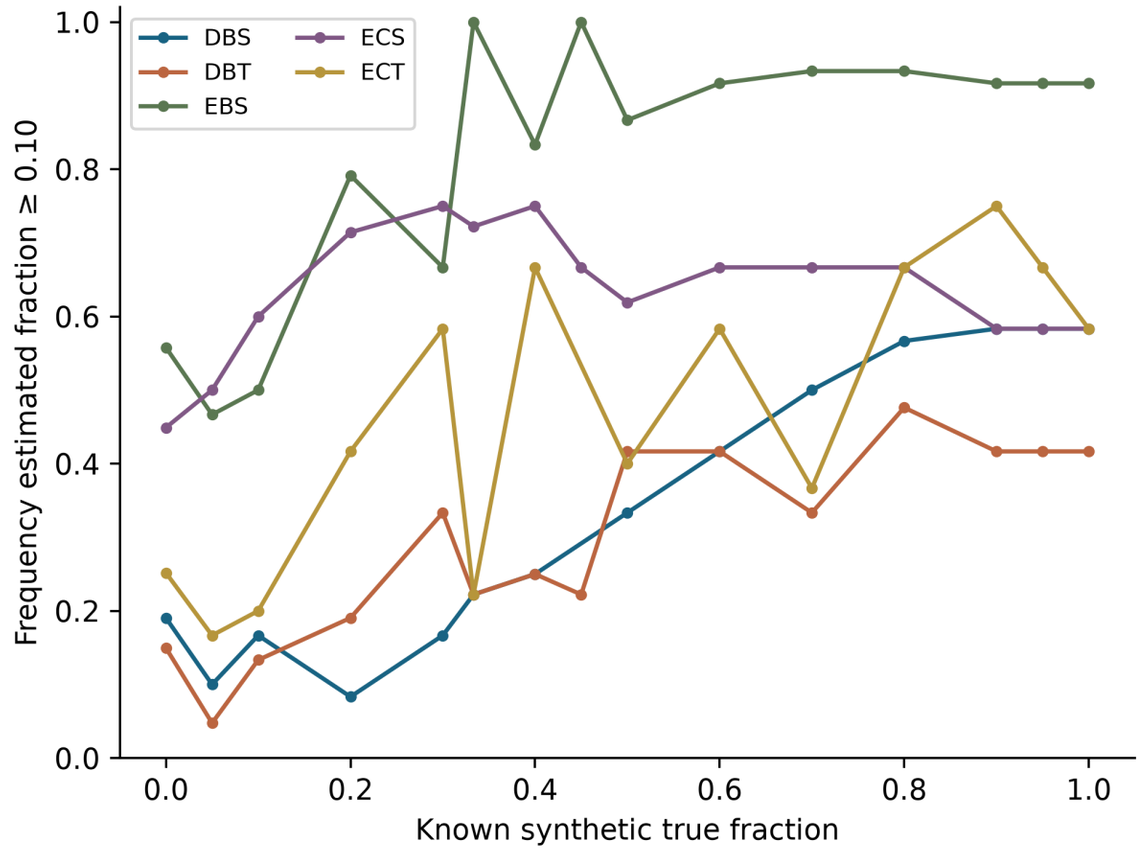
A prototype is a selected observed reference, not a verified pure endmember. Keep all valid raw repeats and their provenance; derive compact references separately. Optimizing a library on the final test set would erase the distinction this experiment is designed to preserve.

Evidence: STAGE5_CLASS_REPRESENTATION.md; class_prototype_coverage.csv. Full source paths and hashes are retained in the report packet.

Figure 19. Synthetic fraction recovery is a harder task than classification

Can the library recover how much of a class is present?





HOW TO READ IT

Left: class/fold-averaged fraction RMSE for selected narrow-coordinate budgets and OLI. Right: fraction of inferred-PFT trials estimated above the 0.10 presence cutoff versus known synthetic fraction. At a true fraction of zero, positive detection is a false detection.

WHAT TO UNDERSTAND

The reused full-spectrum primary inferred-PFT fraction RMSE is about 0.344 in NEON and 0.351 in OLI. Errors and detections vary greatly by class. Some zero-fraction trials are falsely positive, while some truly present classes are missed even at higher fractions.

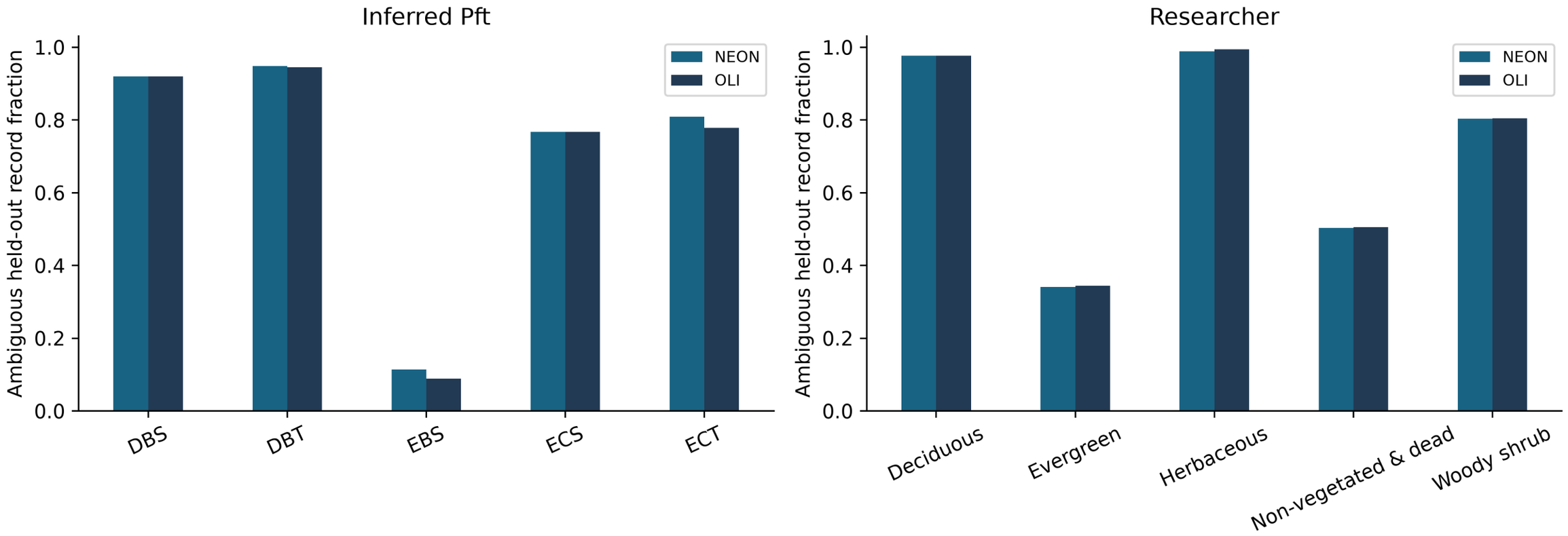
WHAT IT DOES NOT ESTABLISH

Mixtures are constructed from saved observed parents under a controlled linear model; there is no independent field-cover truth. Dependent parent trajectories cannot be treated as independent population detection replicates. No universal 50/80/95% detection limit or calibrated abundance interval is established.

Evidence: STAGE7_MIXTURE_RECOVERY.md; mixture_recovery.csv; detection_limits.csv. Full source paths and hashes are retained in the report packet.

Figure 20. A good spectral fit can have the wrong biological explanation

Is a class spectrum uniquely identifiable against mixtures of other classes?



HOW TO READ IT

Bars show the proportion of held-out class spectra admitting a close fit using candidates from other classes. NEON and OLI are paired. The declared tolerance is 0.01 in a training-normalized analytical metric, not 0.01 absolute reflectance.

WHAT TO UNDERSTAND

Many class spectra have other-class convex alternatives; NEON and OLI show similarly serious ambiguity. EBS has a lower observed rate but is rare. This explains how low residual error can coexist with poor fraction recovery or wrong class allocation.

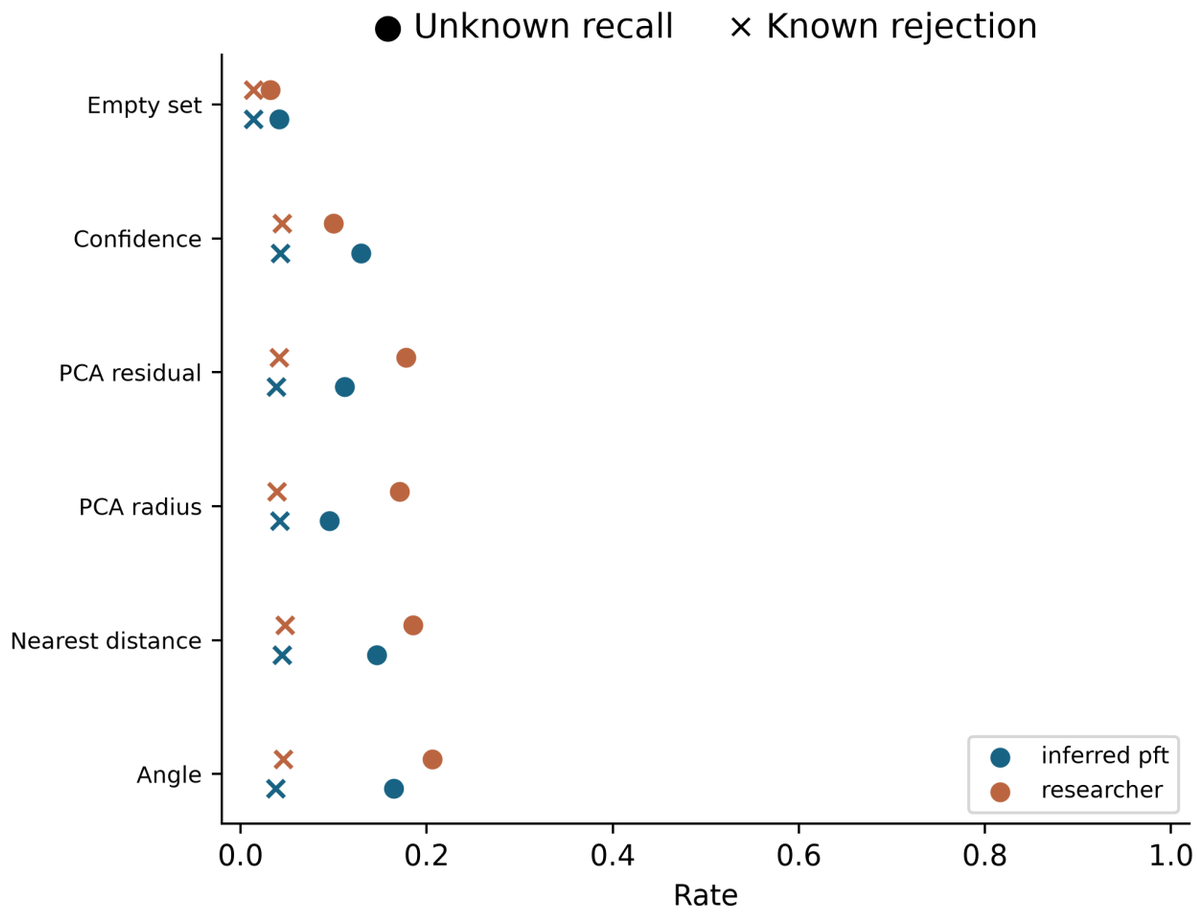
WHAT IT DOES NOT ESTABLISH

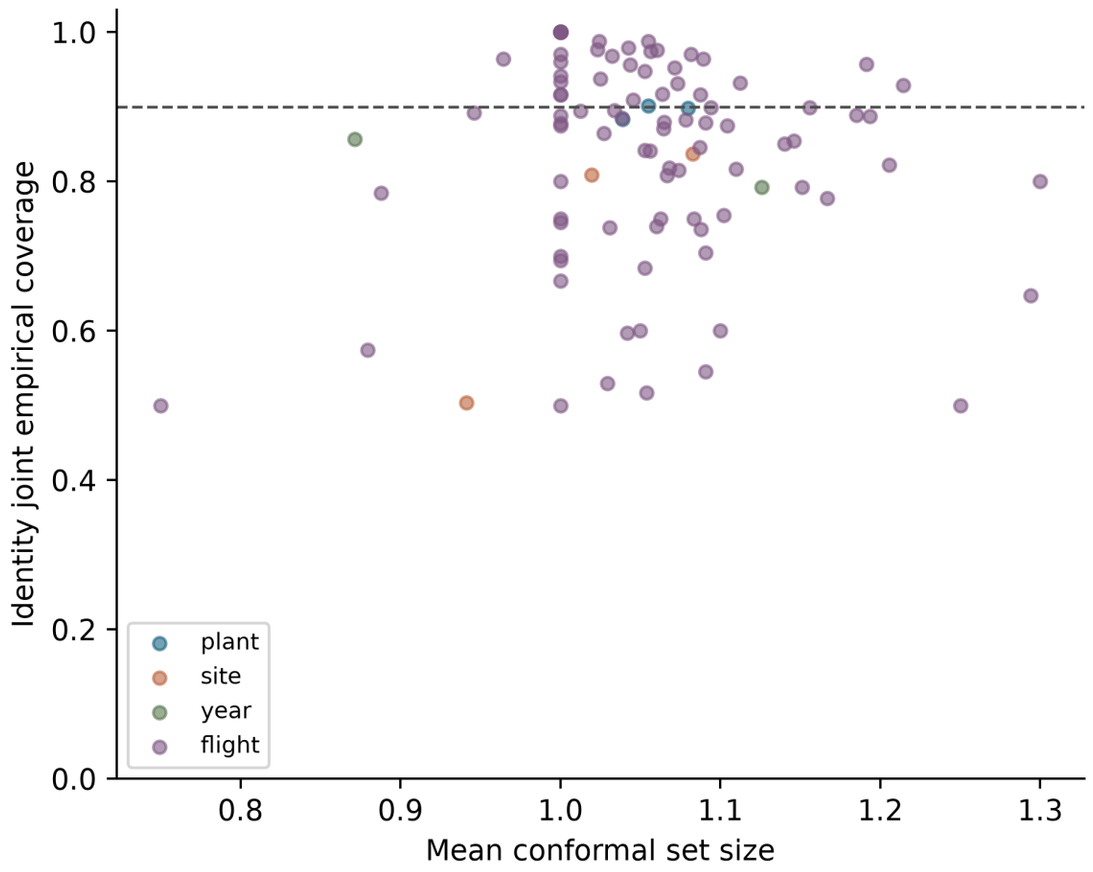
Ambiguity depends on candidate library, normalization, tolerance and parent support; it is not proof that nature makes all PFTs intrinsically indistinguishable. More endmembers can improve approximation while increasing ambiguity. Classification success does not solve this inverse problem.

Evidence: STAGE6_MIXTURE_IDENTIFIABILITY.md; mixture_identifiability.csv. Full source paths and hashes are retained in the report packet.

Figure 21. Confidence is not a reliable detector of unfamiliar biology

Will a model know when a class is absent from its training library?





HOW TO READ IT

Left: circles are unknown-class recall; crosses are rejection of known classes, for thresholds chosen with known calibration only. Higher recall is desirable, but higher known rejection is costly. Right: empirical prediction-set coverage versus mean set size, with 90% nominal coverage marked.

WHAT TO UNDERSTAND

For inferred PFTs, nearest-angle rejection detects only 16.5% of omitted-class challenges while rejecting 3.7% of known cases. Ordinary 90% conformal sets attain roughly 88.4-90.2% joint-identity coverage in new-polygon tests but only 50.3-83.7% at held sites.

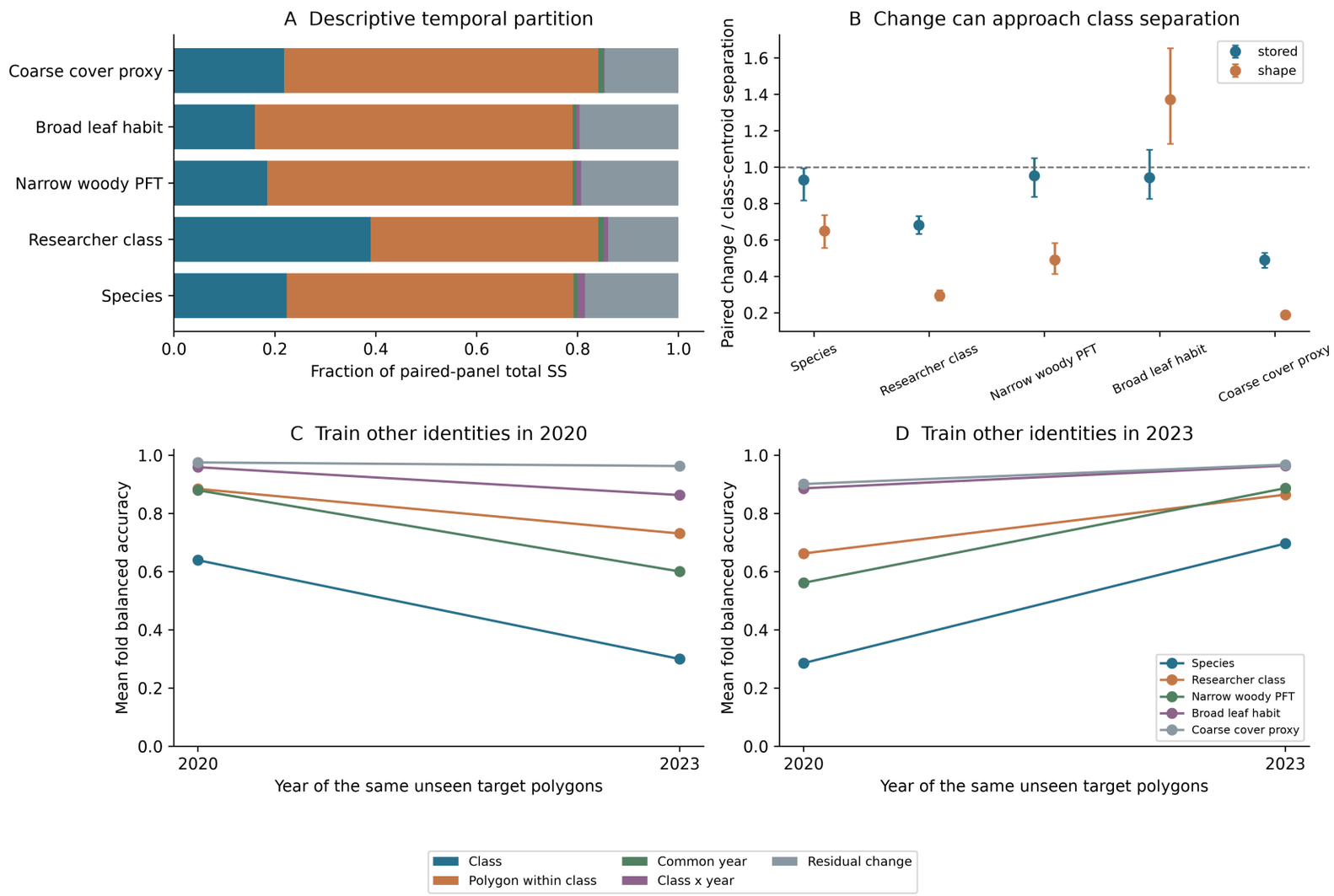
WHAT IT DOES NOT ESTABLISH

Omitted ecological labels are proxies for unknowns, not verified new physical materials. The coverage dots span methods/domains; they are not independent populations. Temperature scaling and small prediction sets do not guarantee reliable uncertainty after geographic shift.

Evidence: STAGE8_UNKNOWN_MATERIALS.md; STAGE9_UNCERTAINTY.md; unknown_material_detection.csv; uncertainty_calibration.csv. Full source paths and hashes are retained in the report packet.

Figure 22. Repeat observations reveal a substantial temporal penalty

Does a label remain recognizable in a different image year?



HOW TO READ IT

A partitions complete-pair spectral sums of squares. B compares paired temporal change with class-centroid separation; a ratio of one means similar magnitudes. C-D train on other identities in one year and evaluate the same unseen target IDs in both years; curves average three folds.

WHAT TO UNDERSTAND

For species, 2020-trained balanced accuracy drops from 63.9% on 2020 targets to 30.0% on their 2023 counterparts. Broad woody leaf habit drops from 95.9% to 86.3%; the coarse-cover proxy is more stable. Temporal change can be comparable to class-centroid separation.

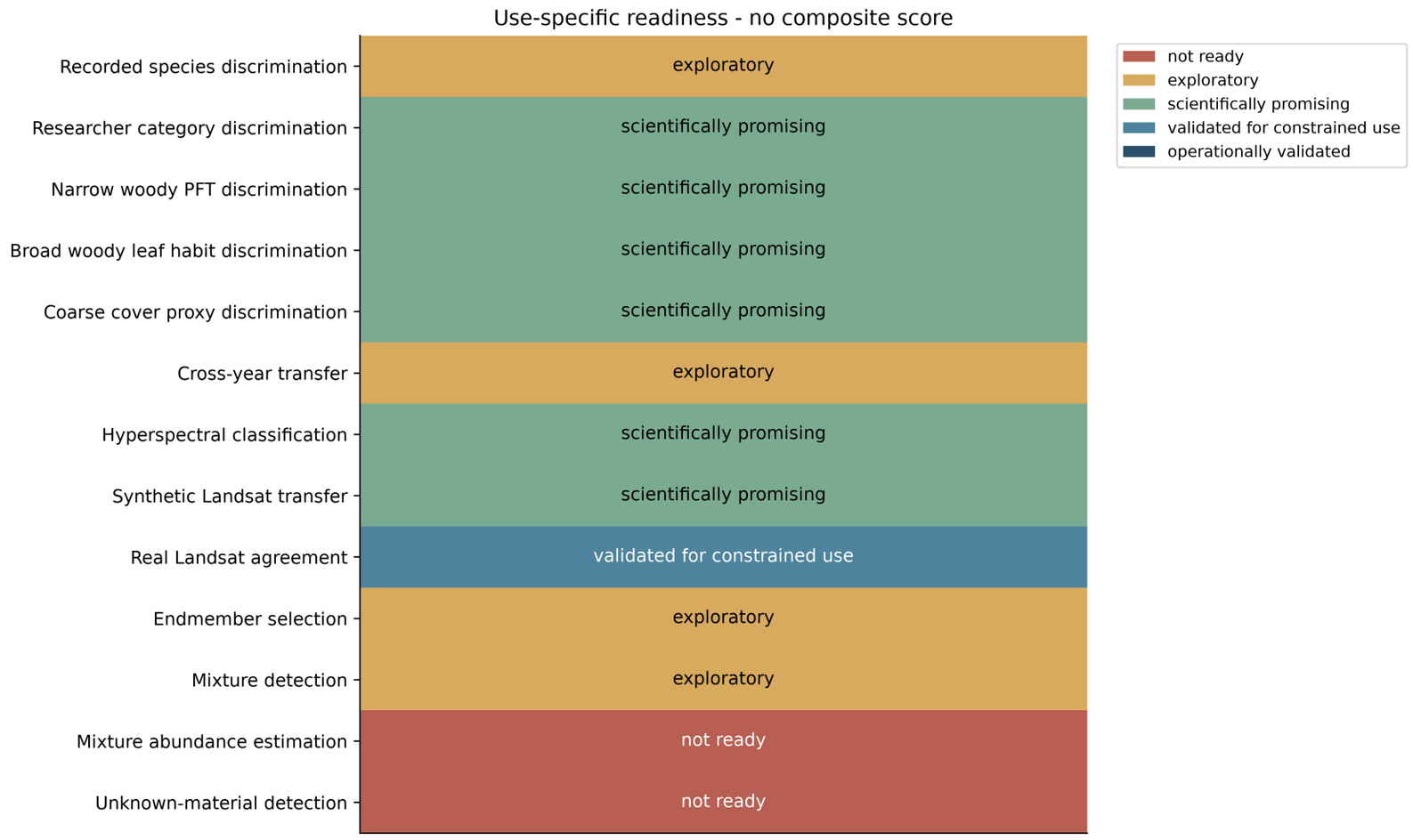
WHAT IT DOES NOT ESTABLISH

All complete pairs are NIWO footprints. Labels come from reused 2023/2024 annotations; there is no independent annual biological verification. The partition is exact descriptive accounting, not causal variance components. Species/classes/populations differ across rows; centroid separation and prediction accuracy are different quantities.

Evidence: paired_temporal_recognition.csv; paired_temporal_accuracy_change.csv; paired_temporal_partition.csv. Full source paths and hashes are retained in the report packet.

Figure 23. Readiness must be decided separately for each intended use

What should the lab use now, and what requires more evidence?



HOW TO READ IT

Each row is a separate use case. Colors are explicit evidence-based readiness categories, not an averaged score. Scientifically promising means conditional evidence merits further use/review; constrained validation applies only within the stated experiment.

WHAT TO UNDERSTAND

Use the library to characterize sampled spectra, compare researcher categories and investigate conditional label recognition. Broad woody leaf habit is the strongest transfer result. The one real-Landsat pilot supports its constrained radiometric comparison. Mixture abundance and dependable unknown detection are not ready.

WHAT IT DOES NOT ESTABLISH

No row is operationally validated. Species and cross-year use remain exploratory. Synthetic sensor transfer is not field validation, and good classification is not evidence of unique unmixing. Future claims need independent labels, physical metadata, balanced domains and independent fractional-cover truth.

Evidence: FINAL_READINESS_MATRIX.csv; DESIGN.md; NEXT_DATA_NEEDED.md. Full source paths and hashes are retained in the report packet.

What we can conclude, and how the library should grow

Use now, within the measured scope

Describe sampled spectral variation; compare matched researcher categories and assigned woody PFTs; benchmark supported-label recognition. Broad woody leaf habit is the most transferable demonstrated classification target. The independent Landsat result is a constrained one-pair radiometric/spatial comparison.

Do not promote to field abundance yet

Pure-spectrum discrimination does not resolve wrong-class mixtures. Unknown rejection is weak, uncertainty loses coverage under site shift, and independent fraction/purity truth is missing. No method is operationally validated by this analysis.

Retain repeats; control their statistical weight

Keep every valid source observation, identity, site, acquisition date and annotation date. Hold complete histories together in validation. Use identity-balanced weighting when the target is an average polygon; make additional class/site weights explicit. Preserve prototypes as derived products, not replacements for the archive.

1. Recover the inexpensive metadata first

Verify original reflectance units/scaling, physical wavelength headers and polygon purity. Resolve MicaSense band identities before ecological inference. A later field survey does not independently verify a polygon's historical biological state.

2. Collect the truth missing from the hard tasks

Measure independent co-located fractional cover and purity at the relevant footprint, including true absences and low positive fractions. Cross site with year and independently update labels, turnover and persistent identity links. More pixels from the same few parents do not replace replication.

3. Replicate real sensor comparisons

Acquire synchronized, physically verified NEON, Landsat and drone observations across sites and seasons, with comparable footprints and independent references. Choose sample sizes from scientific error tolerances and cluster structure. These gaps take priority over another algorithm search.

KEEP THE TERMS STRAIGHT

Glossary for reading and discussing the figures

Polygon / GlobalID

A source-labelled footprint. It can contain one crown, several plants or a mixed patch. It is the identity used for grouped validation, not proof of biological purity.

Polygon/year spectrum

A compact annual summary of source observations for an identity. Two years from the same GlobalID are repeated observations, not two independent individuals.

Researcher categories

Deciduous, Evergreen, Herbaceous, Non-vegetated & dead, Woody shrub. All 3,639 identities are represented. These are recorded cover categories, not a strictly nested physiological hierarchy.

Species-derived woody PFTs

DBS: deciduous broadleaf shrub; DBT: deciduous broadleaf tree; EBS: evergreen broadleaf shrub; ECS: evergreen conifer shrub; ECT: evergreen conifer tree. Assigned subset: 1,944 identities.

Broad leaf habit / coarse cover

Broad leaf habit means deciduous versus evergreen within the assigned woody subset. Coarse cover contrasts vegetation-labelled categories with the heterogeneous Non-vegetated & dead proxy.

NEON / OLI / MicaSense

NEON is the hyperspectral source. OLI is the Landsat sensor representation; synthetic OLI is derived from NEON. MicaSense here denotes exported derived values with unresolved physical coordinates, not an independently observed drone validation.

NIR / SWIR / atmospheric gaps

Near-infrared and shortwave-infrared spectral regions. Excluded atmospheric absorption intervals remain disconnected; lines do not interpolate through those gaps.

Stored units / surface reflectance

Library values retain their stored scale, whose absolute reflectance provenance remains unresolved. The separately verified real-sensor pilot reports unitless surface reflectance. Do not mix their error scales.

Closed set / target coverage

Closed-set prediction evaluates labels supported by training and test eligibility rules. Coverage is the proportion of the intended target cohort retained. High conditional accuracy can coexist with low coverage.

Endmember / prototype

An endmember is a candidate component in a mixture model. A prototype is a representative reference. An observed polygon prototype is not automatically a physically pure endmember.

Where the numbers and figures came from

Exploration and quantile checks

provenance/plot-report/2026-10-13/; provenance/quantile-models/2026-10-13/. Reader-sized exploratory panels are reused from output/pdf/plant_spectral_review/figures/.

Biological contrasts and spectral reduction

provenance/researcher-pft/2026-10-14/; provenance/sensor-pft-transfer/2026-10-14/; provenance/publication-figures/2026-10-14/. Fixed matched comparisons, label systems and metadata gates are preserved.

Generalization, prototypes, mixtures and uncertainty

provenance/spectral-library-analytics/2026-10-14/ and provenance/mesma-readiness/2026-10-14/. Existing models, memberships and canonical numerical experiments are reused.

Independent Landsat and final closure

provenance/real-landsat-validation/2026-10-14/; provenance/scientific-closure/2026-10-15/. Final closure supersedes earlier interpretations where metadata or biological-resolution scope was clarified.

This report packet

provenance/lab-story/2026-10-15/ retains the editable report_source.json, Markdown companion, builders, figure inventory, numerical checks and SHA-256 input/output manifest. All 23 visuals are embedded in this PDF. The packet references existing source data in place; it does not copy the merged polygon table or refit models.

Method reading retained from the earlier audits: Roberts et al. (2017), doi:10.1111/ecog.02881 (structured validation); Cawley & Talbot (2010), JMLR 11:2079-2107 (selection bias); Bioucas-Dias et al. (2012), doi:10.1109/JSTARS.2012.2194696 (unmixing); Angelopoulos & Bates, arXiv:2107.07511 (conformal prediction). See the earlier LITERATURE_METHODS.md files for method adaptations and full links.